Processes Governing the Ablation of Intercepted Snow

Alex C. Cebulski[1], John W. Pomeroy[1]

[1]Centre for Hydrology, University of Saskatchewan, Canmore, Canada,

Key Points:

- Unloading was found to be strongly associated with canopy snow load, wind shear stress, and canopy snowmelt.
- New canopy snow unloading relationships coupled to a physically-based energy balance routine improved canopy snow load predictions.
- The fraction of snowfall sublimated in the canopy differed by up to a factor of three across existing models.

Corresponding author: Alex C. Cebulski, alex.cebulski@usask.ca

Abstract

Interception and ablation of snow in forest canopies significantly influence the quantity, timing, and phase of precipitation that reaches the ground in cold regions forests. Yet current modelling approaches have uncertain transferability across differing climate and forest types. Here, in-situ observations from a needleleaf forest in the Canadian Rockies were utilised to evaluate the theories underpinning existing canopy snow ablation models and develop a novel understanding supporting the development of a new canopy snow ablation model. The observations revealed that canopy snow load, wind shear stress, and canopy snowmelt are strongly associated with unloading; however, air temperature and sublimation are not. A new canopy snow ablation model was developed based on these associations and their impact on the canopy snow energy and mass balance. This model demonstrated improved performance in simulating canopy snow load compared with previous approaches, especially during melt- and wind-dominated ablation events. The improved performance in predicting canopy snow load across a wide range of meteorological conditions, compared to existing models, is due to including a comprehensive representation of the mass and energy balance of intercepted snow. In contrast, all existing canopy snow models were found to omit key processes which limited their accuracy in simulating snow load, its ablation and partitioning to sublimation, melt, drip, and unloading.

1 Introduction

The seasonal snowpack is an essential component of global water resources (Immerzeel et al., 2020; Viviroli et al., 2020) and is increasingly threatened by rapid climate and land cover change (Aubry-Wake & Pomeroy, 2023; Fang & Pomeroy, 2023; López-Moreno et al., 2014; Szczypta et al., 2015). Snow falls in forests over half of the Northern Hemisphere (Kim et al., 2017) and over 23% of the global landmass (Deschamps-Berger et al., 2025), where vegetation cover can significantly alter the quantity, phase, and timing (Ellis et al., 2010; Essery et al., 2003; Mazzotti et al., 2021; Pomeroy & Gray, 1995; Rojas-Heredia et al., 2024; Safa et al., 2021; Sanmiguel-Vallelado et al., 2020) of snowfall reaching the ground. Therefore, forest-snow interactions are crucial to understand for informed ecological, land management, and water resource decision-making. Needleleaf canopies are particularly effective at intercepting snowfall (Cebulski & Pomeroy, 2025b; Hedstrom & Pomeroy, 1998; Pomeroy & Schmidt, 1993; Storck et al., 2002) and expose intercepted snow to increased energy inputs relative to the subcanopy snowpacks, leading to increased rates of melt, unloading, and/or sublimation (Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b; Roesch et al., 2001; Storck et al., 2002). The partitioning of snow to the atmosphere via sublimation (Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b) or to the ground through unloading and meltwater drip (Lumbrazo et al., 2022; Roesch et al., 2001; Storck et al., 2002) is highly sensitive to meteorological conditions and forest structure, contributing to differing process emergence and substantial variability in subcanopy snow accumulation. Coastal humid environments typically exhibit small sublimation losses and a larger influence of unloading and melt (Floyd, 2012; Storck et al., 2002). In these temperate environments, enhanced canopy energy inputs combined with high humidity increase both solid snow unloading and melt of snow intercepted in the canopy, with subsequent drip to beneath the canopy (Lumbrazo et al., 2022; Lundquist et al., 2021; Roesch et al., 2001). High liquid water contents during rapid melt can lubricate snow attached to the canopy and weaken adhesive and cohesive bonds, inducing unloading, much as for wet snow avalanches (Baggi & Schweizer, 2008). Conversely, the colder and drier winters in continental climates typical of boreal forests can drive substantial canopy sublimation losses (e.g., 25–45% of annual snowfall in Essery et al., 2003) in addition to unloading (Essery et al., 2003; Gelfan et al., 2004; Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b, 2002). As a result, reliable models of snow accumulation, sublimation, and runoff generation in forested basins must be predicated on a comprehensive understanding of interception processes (Clark et al., 2015; Essery et al., 2003; Pomeroy et al., 1998a; Verseghy, 2017; Wheater et al., 2022).

Canopy snow models have demonstrated variable degrees of suitability for transfer across different climates and forest types (Essery et al., 2003; Gelfan et al., 2004; Lumbrazo et al., 2022; Lundquist et al., 2021), which has been attributed to limiting the performance in predicting forest snowpacks in hydrological model intercomparisons (Krinner et al., 2018; Rutter et al., 2009). Recent studies have emphasised the importance of distinguishing between initial snow interception and subsequent ablation processes (Cebulski & Pomeroy, 2025a, 2025b). This separation required explicit parameterisations of distinct processes, which improves process representation and is well suited for the modular design of contemporary models, thereby supporting broader applicability across diverse environments and model approaches (Clark et al., 2015; Pomeroy et al., 2022). In addition, Lundquist et al. (2021) and Cebulski & Pomeroy (2025b) demonstrated that canopy snow processes can be more accurately represented without the concept of a maximum canopy snow load, which has been assumed to be an important parameter in existing initial accumulation parameterisations (Andreadis et al., 2009; Hedstrom & Pomeroy, 1998). Moreover, existing canopy snow accumulation parameterisations often implicitly incorporate snow ablation, likely because ablation is included in measurements of initial interception efficiency (Cebulski & Pomeroy, 2025a; Hedstrom & Pomeroy, 1998; Katsushima et al., 2023; Schmidt & Gluns, 1991). For example, the field experiments of Hedstrom & Pomeroy (1998) in the boreal forest showed that interception efficiency declined as the canopy fills with snow, whilst the measurements of Storck et al. (2002) in coastal forests suggested a constant interception efficiency of 0.6, capped by a maximum canopy snow load. Consequently, both the Hedstrom & Pomeroy (1998) and Storck et al.

(2002) parameterisations have significantly lower interception efficiencies—prior to canopy snow ablation—compared to the observations by Cebulski & Pomeroy (2025b). Working in high snowfall, cold continental climate mountain basins, Staines & Pomeroy (2023) and Cebulski & Pomeroy (2025b) show that interception efficiency is best predicted by canopy density alone—consistent with recent rainfall interception parameterisations (e.g., Valante et al., 1997; Zhong et al., 2022)—and that snow load has a small influence on canopy density, challenging the Hedstrom & Pomeroy (1998) and Storck et al. (2002) theories regarding the influence of a maximum snow load on initial snow interception. Together, these studies (Cebulski & Pomeroy, 2025a, 2025b; Lundquist et al., 2021) emphasise the need to revisit existing canopy snow ablation parameterisations.

Unloading of intercepted snow to the ground has previously been found to be associated with air temperature (Katsushima et al., 2023; Roesch et al., 2001; Schmidt & Pomeroy, 1990), ice-bulb temperature—which represents the cooling effect of sublimation on snow temperature—(Ellis et al., 2010; Floyd, 2012), canopy snowmelt rate (Storck et al., 2002), and wind speed (see Supporting Information for example relationships). Hedstrom & Pomeroy (1998) proposed a parameterisation based on canopy snow load and time, with its rate coefficient determined by repeated snow surveys. In addition to the results of empirical studies, deduction based on physical reasoning supports the inclusion of these processes. For example, melt can be expected to accelerate unloading through loss of intercepted snow clump structural integrity by weakening of inter-crystal bonds, and increased liquid water content and hence weakened adhesion of intercepted snow to branches. Wind drag can be expected to accelerate unloading through the shear stress applied to intercepted snow, wind erosion via direct snow entrainment by the atmosphere, and branch swaying (Pomeroy & Gray, 1995). The modulus of elasticity of needleleaf branches increases with air or branch temperature (Schmidt & Pomeroy, 1990), which can increase the likelihood of unloading due to less horizontal branch angles and decreased stiffness to resist swaying due to wind drag, as temperature increases (Raleigh et al., 2022).

Melt of intercepted snow is typically represented by an energy balance approach (Andreadis et al., 2009; Clark et al., 2015; Essery et al., 2025; e.g., Parviainen & Pomeroy, 2000), as a function of air temperature (Roesch et al., 2001), or a function of ice-bulb temperature (Ellis et al., 2010; Floyd, 2012). Sublimation is generally represented using a coupled energy-mass balance approach (e.g., Essery et al., 2003; Parviainen & Pomeroy, 2000; Pomeroy et al., 1998b; Verseghy, 2017). All of these parameterisations, except Verseghy (2017), decrease the latent heat flux from snow intercepted in the canopy as the canopy fills up with snow and its specific surface area decreases, thereby relying on the maximum canopy snow load. Recent measurements of canopy snow load in high snowfall environments have not reached a maximum value (Cebulski & Pomeroy, 2025b; Lundquist et al., 2021; Storck et al., 2002) and the observations exceed previous estimates of maximum canopy snow load (Hedstrom & Pomeroy, 1998; Schmidt & Gluns, 1991).

The merits of using more physically based energy balance methods rather than empirically based functions to calculate snowmelt and sublimation have not been directly assessed through an event-based process investigation. Some models vertically discretise the canopy into multiple layers (e.g., Clark et al., 2015) or into two layers (e.g., Essery et al., 2025) to better capture vertical gradients in the canopy energy balance—for instance, enhanced solar radiation and turbulent exchanges near the canopy top relative to the lower canopy. However, uncertainties in generating forcing data at different canopy heights can limit the implementation and validation of these multi-layer approaches. In contrast, other schemes represent the canopy as a single layer (Parviainen & Pomeroy, 2000; Pomeroy et al., 2022; Verseghy, 2017).

Quantifying individual canopy snow ablation processes, including unloading, wind redistribution, melt, drip, and sublimation remains challenging, even with sophisticated lysimetry (Storck et al., 2002) and eddy-covariance systems (Conway et al., 2018; Harding & Pomeroy, 1996; Harvey et al., 2025). Consequently, some canopy snow ablation parameterisations have been developed using methods, such as observations of above canopy albedo (Bartlett & Verseghy, 2015; Roesch et al., 2001). While these approaches offer useful indices of model performance, they provide limited insight into the accuracy of individual process representations. Weighed tree, weighed snow buckets, and tipping buckets measurements offer more direct process-level observations, but their interpretation can be made uncertain by freeze-thaw cycles and concurrent processes (Floyd, 2012; MacDonald, 2010; Storck et al., 2002). A hybrid diagnostic approach that combines individual process measurements with simulations and employs observations such as canopy snow load from a weighed tree, unloading measured by a weighed subcanopy snow collection buckets, and drip from tipping bucket rain gauges has yet to be applied, but is explored in this study.

The objective of this study is to better understand and predict the influences of meteorology and snow load on intercepted snow ablation. This study specifically examines canopy snow ablation following snowfall and initial interception. Cebulski & Pomeroy (2025b) has addressed processes governing the initial interception of snow in the canopy.

The specific research questions this research aims to address include:

1. How do snow load, air temperature, humidity, wind exposure, canopy snow sublimation, and snowmelt influence the rate of canopy snow unloading?

2. To what extent do current theoretical models of canopy snow ablation align with the findings from detailed in-situ observations in a high snowfall, continental climate subalpine forest?

3. What modifications to existing models are necessary to accurately represent ablation of snow intercepted in the canopy, and what is the resulting improvement in predictive performance from these modifications?

2 Methods

2.1 Study Site

Observations were collected in the Fortress Mountain Research Basin (FMRB), Alberta, Canada, -115° W, 51° N, a continental climate headwater basin in the snowy Canadian Rockies. The site is located 2100 m above sea level on a ridge covered by mature subalpine fir (*Abies lasiocarpa*) and Engelmann spruce (*Picea engelmannii*) forest. Air temperature, humidity, and wind speed were measured at a height of 4.3 m at the Forest Tower Station (Figure 1). Shear stress was calculated using EddyPro software (LI-COR Biosciences) based on high-frequency wind measurements from a CSAT3 three-dimensional sonic anemometer (Campbell Scientific) installed at a 3.0 m height on Forest Tower station. The CSAT3 was occasionally snow-covered during the analysis period, and thus to provide a continuous record of shear stress, a linear relationship was established between shear stress derived from the CSAT3 and the square of wind speed measured at a 4.3 m height on the Forest Tower station ($R^2 = 0.71$, $p < 0.05$). This relationship was then used to fill gaps in shear stress measurements during periods when the CSAT3 was snow-covered. The precipitation rate was measured by an Alter-shielded OTT Pluvio weighing precipitation gauge with an orifice 2.6 m above the ground at the adjacent Powerline Station (Figure 1). Incoming and outgoing solar radiation were measured by a heated and ventilated Kipp & Zonen CNR4 4-Component Net Radiometer installed 3.27 m above the ground on Fortress Ridge Station, 2.0 km to the northwest of Forest Tower station (50.8° N, 115.2° W). This wind-exposed site was selected to reduce snow accumulation on the radiometers. Three subcanopy snow collection buckets, consisting of plastic horse-watering troughs with an opening of 0.9 m^2 and depth of 20 cm suspended from a load cell, were installed just above the subcanopy snow survey to measure subcanopy snow accumulation. To isolate subcanopy measurements of canopy snow unloading from throughfall, 15-minute intervals with no atmospheric precipitation were selected based on precipitation measurements at Powerline. A weighed tree lysimeter (subalpine fir) suspended from a load cell (Artech S-Type 20210-100) measured the weight of canopy snow load (kg) and was scaled to an areal estimate of snow load (mm) using snow surveys as in Pomeroy & Schmidt (1993). The weighed tree and time-lapse imagery were also used to confirm the absence of atmospheric precipitation and to identify periods when snow was intercepted in the canopy and ablation was possible. Four tipping bucket rain gauges—3 Texas Electronics TR-525M and 1 Hyquest Solutions TB4MM—were installed along a 15 m transect adjacent to the dense canopy subcanopy weighed snow bucket to quantify sub-canopy drip from melting intercepted snow. Additional details on this study site and the meteorological and weighed snow bucket measurements have been described in Cebulski & Pomeroy (2025b).

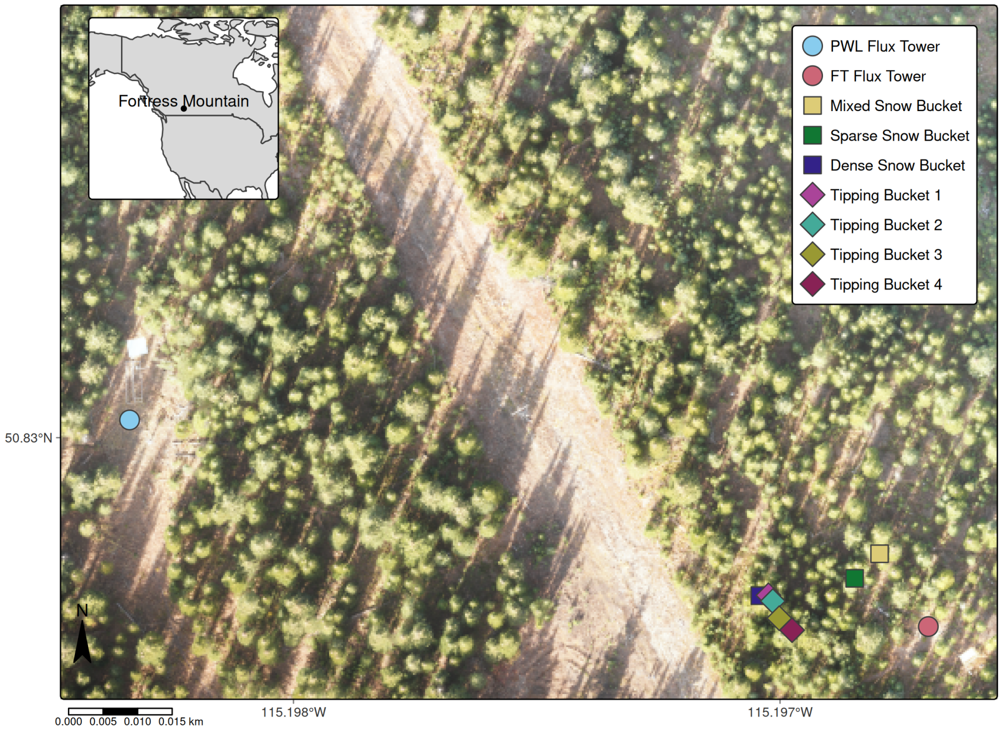

Figure 1: Map showing the location of flux towers, weighed tree, weighed snow buckets, and tipping bucket rain gauges. The inset map on the upper left shows the regional location of Fortress Mountain Research basin. Flux towers are denoted by circles, snow buckets by squares, and tipping bucket rain gauges by diamonds.

2.2 The Cold Regions Hydrological Model Platform

The Cold Regions Hydrological Model Platform (CRHM) was used to implement calculations of the canopy snow energy and mass budget. A full description of the CRHM platform is described in Pomeroy et al. (2022), and the source code is available at https://github.com/srlabUsask/crhmcode. The climate forcing data used to run CRHM were provided from station-based fifteen-minute interval measurements of air temperature, relative humidity, wind speed, precipitation, and incoming solar radiation from the station measurements. CRHM incorporates a flexible modular design allowing the user to select various parameterisations that represent hydrological processes. The phase of atmospheric precipitation was determined from the energy balance of falling hydrometeors (Harder & Pomeroy, 2013). A new CRHM module was created here to simulate the coupled mass and energy balance of snow intercepted in the canopy. The energy balance is described in detail in Section 2.5, and the updates to the mass balance through revisions to the canopy snow unloading functions are presented in Section 3.1.1 and Section 3.1.2. Due to uncertainties in estimating model forcing data and measuring canopy snow ablation at varying canopy heights, a single-layer canopy snow model was implemented in CRHM.

2.3 Canopy Snow Mass Balance

The following mass balance, as described in Cebulski & Pomeroy (2025a) was implemented as a module in CRHM to track the state of canopy snow load (L, mm) over time:

$$\frac{dL}{dt} = [q_{sf} - q_{tf} + q_{ros}] - q_{unld} - q_{drip} - q^{veg}_{wind} - q^{veg}_{sub} \tag{1}$$

where $\frac{dL}{dt}$ is the change in canopy snow load over time, (mm s^{-1}), q_{sf} is the snowfall rate (mm s^{-1}), q_{tf} is the throughfall rate (mm s^{-1}), q_{ros} is the rate of rainfall falling on snow intercepted in the canopy (mm s^{-1}), q_{unld} is the unloading rate due to melt (q^{melt}_{unld}, mm s^{-1}) and due to shear stress on snow, wind erosion, branch movement, structural degradation, bond weakening and increased elasticity of branches and other non-melt related processes (q^{dry}_{unld}, mm s^{-1}), q_{drip} is the canopy snow drip rate (mm s^{-1}) resulting from canopy snowmelt (q^{veg}_{melt}, mm s^{-1}), q^{veg}_{wind} is the wind transport rate in or out of the control volume (mm s^{-1}), and q^{veg}_{sub} is the intercepted snow sublimation rate (mm s^{-1}) which also includes evaporation of liquid water in the canopy. Figure 1 in Cebulski & Pomeroy (2025a) presents a visual representation of this mass balance.

2.4 Mass Balance Parameterisations

In addition to the updated canopy snow model presented in this study, also referred to as CP25 (described in Section 3.1.1 and Section 3.1.2), three other canopy snow models were implemented in as modules in CRHM following previous studies by Ellis et al. (2010) and Floyd (2012) (E10), Roesch et al. (2001) (R01), and Andreadis et al. (2009), who built on observations by Storck et al. (2002) (SA09) (Table 1). The E10 model includes canopy snow sublimation as described in Pomeroy et al. (1998b), dry-snow unloading as a function of canopy snow load from Hedstrom & Pomeroy (1998) with modifications described in Ellis et al. (2010) and Floyd (2012) to handle canopy snow melt and drip processes using an ice-bulb temperature threshold. The R01 model represents dry-snow unloading, melt, and drip using linear functions of wind speed and air temperature, while sublimation is simulated using the parameterisation from Pomeroy et al. (1998b). The SA09 model unloads snow as a fraction of the canopy snowmelt rate, following observations by Storck et al. (2002), and does not include dry-snow unloading. The snowmelt rate from Equation 3 was used to calculate the unloading rate for the Andreadis et al. (2009) unloading and thus differs from the energy balance routine described in their study. Canopy snow sublimation in SA09 was represented using the Essery et al. (2003) parameterisation (Equation 10) as implemented in CP25.

Table 1: Description of canopy snow unloading parameterisations.

Name	Study	Dry-Snow Unloading (q_{unld}^{dry})	Melt Induced Unloading (q_{unld}^{melt})
CP25	This study	f(snow load, wind shear stress), Eq. 12	f(snow load, canopy snowmelt rate), Eq. 14
E10	Ellis et al., (2010); Hedstrom & Pomeroy (1998)	f(snow load), Eq. 5 in Hedstrom & Pomeroy (1998)	f(ice-bulb temp.), see description following Eq. 28 in Cebulski & Pomeroy (2025a)
SA09	Storck et al., (2002); Andreadis et al., (2009)	NA	f(canopy snowmelt rate), Eq. 33 in Andreadis et al., (2009)
R01	Roesch et al., (2001)	f(snow load, wind speed), Eq. 15 in Roesch et al., (2001)	f(snow load, air temp.), Eq. 14 in Roesch et al., (2001)

The retention of canopy snow meltwater differs across the four canopy snow models implemented in CRHM. For E10, liquid meltwater is not retained in the canopy and immediately drips before it can evaporate. The R01 model also does not account for evaporation of liquid meltwater, as canopy snow is not explicitly separated into solid snow unloading and melt/drip. The canopy liquid water storage capacity (L_{max}^{liq}, mm) from SA09 and CP25 was calculated as:

$$L_{max}^{liq} = Lh + cLAI \tag{2}$$

where h is the liquid meltwater holding capacity that the canopy can retain against gravity and n is a storage constant of the vegetation elements. Andreadis et al. (2009) estimated h as 0.035 (-) and n as e^{-4}. In this study, h was set to 0.01 (-), and c was defined as $C_c \quad \cdot \quad 0.2$. A one-sided effective plant (leaf) area index (LAI) was used, including the clumping factor to account for needleleaf canopy structures, following the approach of Ellis et al. (2010) for rainfall interception.

2.5 Canopy Snow Energy Balance

The CP25 and SA09 canopy snow models implemented the following energy balance approach to calculate the energy available for melting snow intercepted in the canopy (Q_{melt}, W m^{-2}) and to track the canopy snow temperature over time ($\frac{\Delta T_{vs}}{\Delta t}$, K s^{-1}). The energy balance is expressed as:

$$Q_{melt} = Q_{sw} + Q_{lw} + Q_p + Q_h + Q_l - [C_p^{ice} L \frac{\Delta T_{vs}}{\Delta t}] \tag{3}$$

where Q_{sw} and Q_{lw} (W m^{-2}) are the net shortwave and longwave radiation heat fluxes to the canopy snow, Q_p (W m^{-2}), is the advective energy rate, and Q_l and Q_h (W m^{-2}), are the turbulent fluxes of latent heat and sensible heat respectively (positive towards canopy snow), and C_p^{ice} (J kg^{-1} K^{-1}) is the specific heat capacity of ice. Figure 2 in Cebulski & Pomeroy (2025a) illustrates this energy balance.

2.6 Energy Balance Parameterisations

The E10 and R01 parameterisations relied on physically guided relationships to simulate sublimation and melt of intercepted and are described in full detail in their respective articles and summarised in Cebulski & Pomeroy (2025a). The following section describes the energy balance parameterisations implemented in the CP25 and SA09 models.

Q_{sw} was determined as:

$$Q_{sw} = Q_{sw}^{in} \cdot (1 - \alpha_s) \cdot \tau_{50}^{veg} \tag{4}$$

where Q_{sw}^{in} is the downwelling shortwave radiation (W m^{-2}), α_s is the albedo of snow intercepted in the canopy (-), τ_{50}^{veg} (-) is the canopy transmittance to Q_{sw}^{in}. Equation 10 from Pomeroy et al. (2009) was used to determine τ_{50}^{veg}, using half of the leaf area index (LAI) based on studies by Weiskittel et al. (2009) and Kesselring et al. (2024), which indicate that approximately 50% of the total leaf area is concentrated in the upper half of coniferous canopies. Q_{sw} provides an approximation of the net shortwave radiation to all snow intercepted in the canopy and is a simplification from using a partial differential equation to determine the radiation incident on individual height layers within the canopy. Upwelling shortwave radiation reflected off the surface snowpack is considered a negligible contribution to the snow intercepted in the canopy, as it is primarily blocked by vegetation elements underlying the canopy snow (Pomeroy et al., 2009).

Q_{lw} was approximated as:

$$Q_{lw} = \downarrow Q_{lw}^{atm} + \uparrow Q_{lw}^{veg,out} - Q_{lw}^{vs,out} \tag{5}$$

where Q_{lw}^{atm} is the downwelling longwave radiation from the atmosphere (W m^{-2}), $Q_{lw}^{veg,out}$ is the longwave radiation upwelling from vegetation elements underlying snow intercepted in the canopy (W m^{-2}), and $Q_{lw}^{vs,out}$ (W m^{-2}) is the outgoing longwave radiation from the top and bottom of the canopy snow layer calculated as:

$$Q_{lw}^{vs,out} = 2\epsilon_s \sigma T_{vs}^4 \tag{6}$$

where ϵ_s is the emissivity (-) of snow taken as 0.99 and σ is the Stefan-Boltzmann (5.67e^{-8} W m^{-1} K^{-4}). Q_{lw}^{atm} was approximated in this study as in Sicart et al. (2006) to represent the influence of atmospheric moisture and clouds on emissivity. $Q_{lw}^{veg,out}$ was calculated with the assumption that canopy elements are in equilibrium with the air temperature plus any increase in vegetation temperature from the extinction of Q_{sw}^{in} in the canopy (Pomeroy et al., 2009, Eq. 4).

Q_p was calculated as:

$$Q_p = [C_p^{liq} m_r (T_r - T_{vs}) + C_p^{ice} m_s (T_s - T_{vs})]/\Delta t \tag{7}$$

where C_p^{liq} is the specific heat capacity of liquid water (J kg^{-1} K^{-1}), m_r is the specific mass of liquid water in precipitation (mm), T_r is the rainfall temperature (K), m_s is the specific mass of snow in precipitation (mm), and T_s is the snowfall temperature (K).

Q_h was calculated as:

$$Q_h = \frac{\rho_a}{r_a} C_p^{air} (T_a - T_{vs}) \tag{8}$$

where ρ_a is the air density (kg m^{-3}), C_p^{air} is the specific heat capacity of air (J kg^{-1} K^{-1}), T_a is the air temperature, and r_a is the aerodynamic resistance (s m^{-1}) which was approximated from Equation 4 in Allan et al. (1998) as:

$$r_a = \frac{\log(\frac{z_T - d_0}{z_0}) \log(\frac{z_u - d_0}{z_0})}{\kappa^2 u_z} \tag{9}$$

where z_T is the height of temperature measurement (m), d_0 is the displacement height (m) which was approximated as 2/3$^{\text{rd}}$ the mean canopy height, z_0 is the roughness length (m) which was approximated as 1/10$^{\text{th}}$ of the mean canopy height, z_u is the wind speed measurement height (m), κ is von Kármán's constant, 0.41 (-), and u_z is the wind speed measurement at z_u (m s^{-1}).

Q_l was calculated as:

$$Q_l = \frac{\rho_a}{r_i + r_a} (q_a(T_a) - q_{vs}(T_{vs})) \tag{10}$$

where r_i is a resistance for transport of moisture from intercepted snow to the canopy air space (Eq. 28 in Essery et al., 2003), $q_a(T_a)$ and $q_{vs}(T_{vs})$ are the specific humidity (-) at the air temperature and canopy snow temperature, respectively. r_i was calculated following the concept of how full the canopy is with snow as introduced by Pomeroy & Schmidt (1993), with modifications to incorporate a larger maximum canopy snow load capacity of 50 mm based on observations by Storck et al. (2002), Floyd (2012), and Cebulski & Pomeroy (2025b).

The above sensible and latent heat flux equations assume neutral atmospheric stability conditions, which is supported by the uncertainty in stability corrections in forest canopies (Conway et al., 2018) and mountain environments in winter (Helgason & Pomeroy, 2012a). Solving Equation 3 requires an iterative solution to determine ΔT_{vs} and the remaining terms, which are also a function of T_{vs}.

2.7 Influence of Predictive Variables on Unloading

The following hypotheses were tested to determine dominant predictors of unloading:

a. Melt promotes unloading through loss of structural integrity, particle bond weakening, and lubrication of intercepted snow.
b. Sublimation promotes unloading via structural degradation and bond weakening of intercepted snow.
c. Wind drag promotes unloading through shear stress applied to intercepted snow, wind erosion through direct entrainment in the atmosphere of intercepted snow, and branch movement.
d. Increasing air temperature promotes unloading by increasing the elasticity of branches and its association with melt and/or sublimation.

Distributions of air temperature, wind speed, relative humidity, canopy snow unloading, melt, and sublimation were compared between periods with simulated canopy snow melt and periods without melt. A canopy snowmelt threshold of 0 $\mathrm{mm\,hr^{-1}}$ was used to define non-melt periods, with values greater than 0 $\mathrm{mm\,hr^{-1}}$ indicating melt periods. These comparisons were used to evaluate whether the samples represented the same underlying distribution, thereby determining whether the hypotheses and statistical relationships should be analysed separately for melt and non-melt periods. A two sample non-parametric Wilcoxon signed-rank test was used to evaluate whether distributions of the variables between the two periods differ by a location shift of zero. In addition, the non-parametric Kolmogorov-Smirnov test was applied to compare the two samples as a whole and assess whether they were drawn from the same continuous distribution.

The subcanopy weighed snow bucket unloading measurements had a high relative instrument error due to the relatively small accumulation of unloaded snow over the 15-minute intervals. To improve instrument accuracy, whilst maintaining consistency of the unloading measurements with the independent variables, and minimise issues related to temporal autocorrelation, the 15-minute interval measurements of unloading were aggregated into bins defined by ranges of the predictor variables. Independent variable bins that had less than 0.1 mm of accumulated snow were removed, resulting in a mean instrument error of +/- 2% for the remaining bins. Air temperature and wind speed were measured at Forest Tower station, canopy snow load from the weighed tree lysimeter (scaled to the canopy of each respective subcanopy weighed snow bucket), and canopy snowmelt and sublimation were simulated using CRHM with the CP25 model as described in the previous section.

Predictive relationships of canopy snow unloading were formulated using differing combinations of predictor variables, including canopy snow load, air temperature, ice-bulb temperature, ice-bulb temperature depression (air temperature subtracted by ice-bulb temperature), wind speed, shear stress, and canopy snowmelt. These relationships were evaluated using the mean bias, root-mean-square error (RMSE), coefficient of determination (R^2), and coefficient of agreement (d) (Willmott, 1982). Linear relationships that did not include an intercept term had R^2 values adjusted following (Kozak & Kozak, 1995). Binning unloading across the different predictor variables produced distinct unloading distributions for each predictor, which precluded the use of the Akaike Information Criterion metric for evaluating model performance. Linear relationships were fit using ordinary least squares regression, and nonlinear relationships using nonlinear least squares regression. Multicollinearity was evaluated via Variance Inflation Factors, Pearson's correlation, and non-linear correlations amongst predictors were assessed using Spearman's rank correlation coefficient. Relationships failing the homoscedasticity test (Breusch-Pagan $p < 0.05$) were modelled using generalised least squares (GLS) regression, which accounts for unequal residual variances by appropriately weighting observations. P-values are not reported for GLS coefficients, as heteroscedasticity violates the t-distribution assumptions underlying standard significance tests.

2.7.1 Melt Driven Unloading

A mass balance approach was incorporated to determine the unloading rate, q_{unld}^{melt} during melt periods. The effect of the canopy snowmelt rate (q_{melt}^{veg}) on unloading was then assessed by fitting a linear model using an ordinary least squares regression. Since direct measurements of canopy snow unloading are challenging to obtain independently of canopy snowmelt drainage (Storck et al., 2002), the mass balance introduced in Equation 1 was incorporated to determine q_{unld}^{melt} as a residual. During intervals without precipitation or q_{wind}^{veg} Equation 1 was simplified and rearranged to:

$$q_{unld}^{melt} = -\frac{dL}{dt} - q_{drip} - q_{sub}^{veg} \tag{11}$$

While some components of the canopy snow mass balance can be measured directly, such as $\frac{\Delta L}{\Delta t}$ with the weighed tree, sublimation of canopy snow is more difficult to quantify directly especially in forested mountain environments (Conway et al., 2018; Harding & Pomeroy, 1996; Helgason & Pomeroy, 2012b; Parviainen & Pomeroy, 2000) and thus q_{sub}^{veg} was simulated in this study in CRHM using Equation 10 as described in Essery et al. (2003). q_{drip} was measured where possible using rain gauges; however, freezing of liquid water in the tipping bucket mechanisms limited the ability to measure q_{drip} reliably. Thus, q_{drip} was also estimated using simulations of canopy snowmelt (q_{melt}^{veg}) in CRHM as in Equation 3, with storage limited by the canopy liquid water holding capacity and drainage of excess water.

2.8 Model Evaluation

Performance of the CP25, E10, R01, and SA09 models was assessed using the following error metrics: mean bias, root-mean-squared error (RMSE), Nash-Sutcliffe Efficiency (NSE, Nash & Sutcliffe, 1970), and Kling-Gupta Efficiency (KGE, Clark et al., 2021; Gupta et al., 2009). Observed canopy snow ablation from the weighed tree was compared against model simulations to compute these error metrics at an hourly timestep. To quantify the sampling uncertainty related to the potential influence of outlier events, a bootstrap resampling procedure with 5000 replicates was applied following a similar approach to Clark et al. (2021), using differing combinations of entire canopy snow ablation events which were resampled as discrete data blocks.

3 Results

3.1 Unloading Relationships

Distributions of air temperature, wind speed, unloading rate, sublimation rate, and canopy snow snowmelt rate differed significantly between melt and non-melt ablation periods based on the non-parametric Wilcoxon signed-rank test ($p < 0.05$, Figure 2). Relative humidity did not differ significantly based on the Wilcoxon test ($p > 0.05$). The non-parametric Kolmogorov-Smirnov test indicated that all predictors originated from differing distributions ($p < 0.05$). These statistical differences, together with the distinct physical processes operating during canopy snow melt and non-melt conditions, supported treating the two event types separately as in subsequent analyses (see Supporting Information for full test statistics).

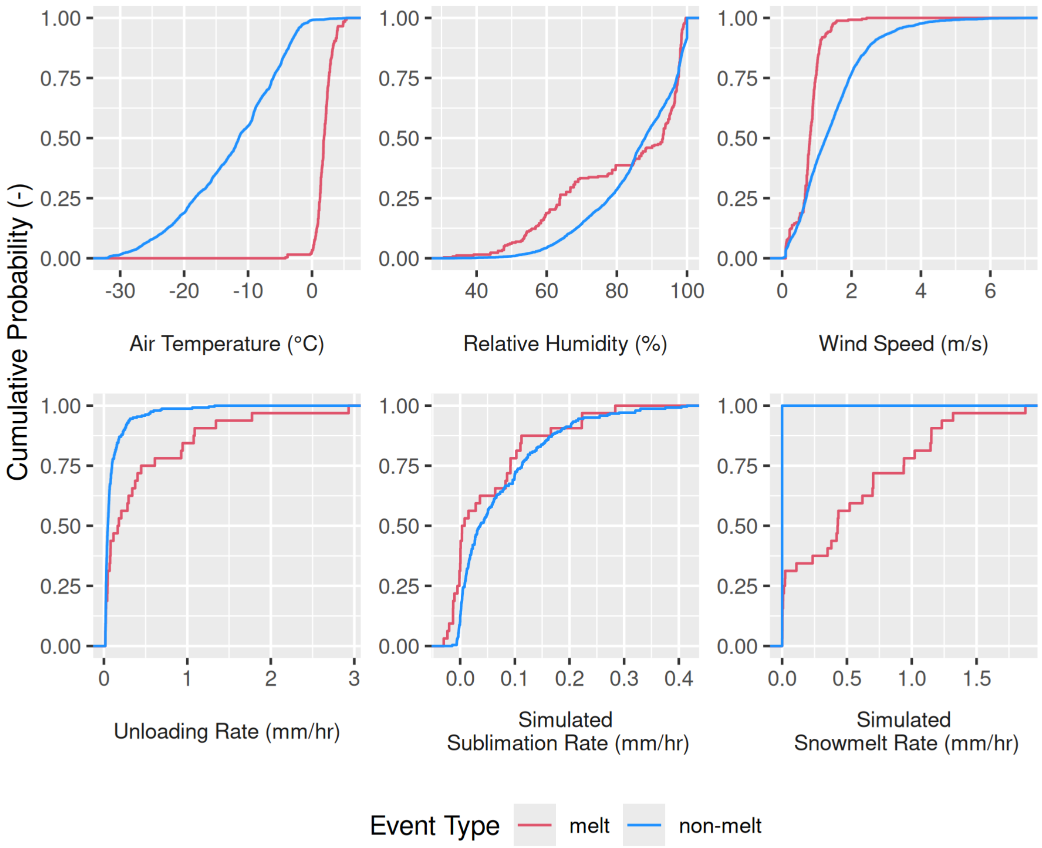

Figure 2: Cumulative probability distributions of observed air temperature, relative humidity, wind speed, and unloading rate, along with simulated canopy snow sublimation and snowmelt rates, for unloading events occurring under canopy snow melt and non-melt conditions. Note: Unloading data has been aggregated over 6-hour time intervals.

3.1.1 Dry-Snow Unloading

A linear relationship with shear stress and an exponential relationship with wind speed, each combined with snow load, produced the lowest errors when predicting unloading during the non-melt periods. The RMSE values were 0.155 mm hr^{-1} and 0.11 mm hr^{-1} for the shear stress and wind speed relationships, respectively, with corresponding R^2 values of 0.58 and 0.54 (Table 2). In addition to model approximation error, uncertainty arose from instrument limitations and from unloading processes unrelated to wind. The shear stress relationship fitted using ordinary least squares violated the homoscedasticity assumption (Breusch-Pagan test $p < 0.05$; see Supporting Information section), with residual variance increasing for higher unloading rates. To address this issue, a generalised least squares regression was applied, which accounts for differing residual variances by appropriately weighting observations.

Additional relationships that had differing combinations of canopy snow load, shear stress, sublimation, air temperature, or ice-bulb temperature depression all produced higher RMSE values and lower R^2 and d statistics than the wind speed and shear stress relationships (Table 2). Furthermore, a model that included snow load, shear stress, and air temperature was statistically insignificant ($p > 0.05$). Several predictor combinations—specifically shear stress with wind speed, air temperature with ice-bulb temperature depression, and sublimation rate with ice-bulb temperature depression—were strongly correlated (Pearson or Spearman correlation coefficients > 0.7) and therefore were not used together in any model evaluation. The full correlation matrices are provided in the Supporting Information.

Although the snow load and wind speed predictive relationship had the smallest RMSE, the snow load and shear stress relationship had higher R^2 and d statistics and lower mean bias (Table 2). This, coupled with its better physical representation of kinetic energy, suggests that shear stress is preferred over wind speed as a predictor of dry-snow unloading (Table 2). Accordingly, the dry unloading rate, q_{unld}^{dry}, was represented as a linear function of snow load (L) and shear stress:

$$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a \tag{12}$$

where τ_{mid} is the shear stress at mid-canopy height (N m^{-2}) and a is a fitting constant.

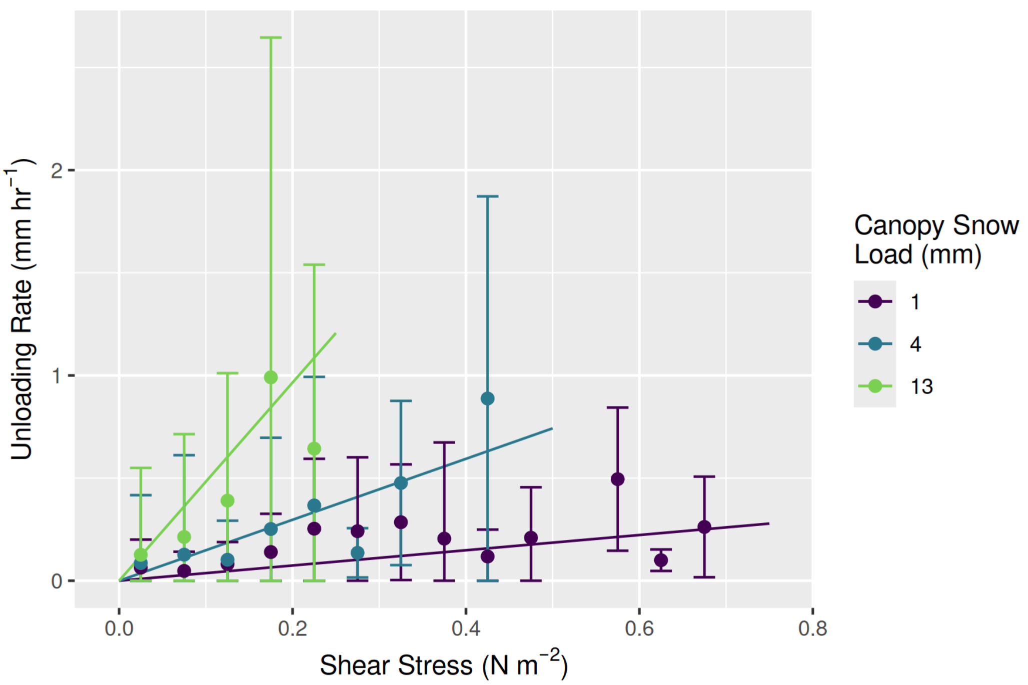

Figure 3: The canopy snow unloading rate measured by the subcanopy weighed snow buckets versus shear stress during periods without canopy snowmelt. The dots represent mean unloading rates within bins of shear stress for three canopy snow load levels; error bars indicate +/- 1 standard deviation. The fitted lines show predictions from Equation 12. The canopy snow load corresponds to the midpoint of each bin (0–2, 2–6, 6–20 mm).

Table 2: Regression statistics for dry-snow unloading (q_{unld}^{dry}, mm hr^{-1}) with various predictor combinations derived using ordinary least squares (OLS), non-linear least squares (NLS), and generalised least squares (GLS) regression. Error metrics include mean bias (MB), root-mean-squared error (RMSE), coefficient of determination (R^2), and index of agreement (d). Row names (a) and (b) indicate coefficients, with p(a) and p(b) as their p-values. Equation variables include: canopy snow load (L, mm), mid canopy shear stress (τ_{mid}, N m^{-2}), mid canopy wind speed (u_{mid}, m s^{-1}), canopy snow sublimation rate (q_{subl}^{veg} mm hr^{-1}), air temperature (T_a, °C), and ice-bulb temperature (T_i, °C).

	Snow Load	Shear Stress	Wind Speed	Sublimation	Shear Stress, Air Temperature	Shear Stress, Ice-Bulb dep.	Shear Stress, Sublimation
Fit	GLS	GLS	NLS	OLS	OLS	GLS	GLS
Eq	$q_{unld}^{dry} = L \cdot a$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot a$	$q_{unld}^{dry} = L \cdot u_{mid} \cdot a \cdot e^{b \cdot u_{mid}}$	$q_{unld}^{dry} = L \cdot q_{subl}^{veg} \cdot a$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot T_a \cdot a$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot (T_a - T_i) \cdot a$	$q_{unld}^{dry} = L \cdot \tau_{mid} \cdot q_{subl}^{veg} \cdot a$
MB	0.001	0.008	0.048	0.071	0.085	0.116	0.043
RMSE	0.159	0.155	0.11	0.157	0.227	0.331	0.255
R^2	0.24	0.58	0.54	0.16	-0.01	-0.16	-0.41
d	0.53	0.68	0.67	0.49	0.51	0.5	0.47
a	2.28×10^{-02}	3.71×10^{-01}	4.62×10^{-03}	1.55×10^{-01}	-2.11×10^{-02}	4.08×10^{-01}	$2.72 \times 10^{+00}$
p(a)	NA for GLS	NA for GLS	$p < 0.05$	$p < 0.05$	$p < 0.05$	NA for GLS	NA for GLS
b	NA	NA	3.93×10^{-01}	NA	NA	NA	NA
p(b)	NA	NA	$p < 0.05$	NA	NA	NA	NA

3.1.2 Melt Driven Unloading

Over the melt periods a linear relationship including canopy snow load and simulated canopy snowmelt rate had the strongest relationship with unloading and drip measured from the subcanopy weighed snow buckets ($p < 0.05$, $R^2 = 0.32$, RMSE = 0.501 mm hr^{-1}) (Table 3). The temperature-based variables predicted unloading poorly ($p < 0.05$, $R^2 \leq 0.1$, RMSE ≈ 0.5 mm hr^{-1}). The superior performance of snowmelt rate relative to predictors based on air temperature and ice-bulb temperature reflects its stronger physical basis, as it represents the energy available for melt from all components of the energy balance.

Incorporating shear stress as a multiplicative term in the canopy snowmelt unloading formulation did not improve model performance (Table 3). This may be because wind speeds during canopy snowmelt were generally calm during the melt periods (mostly < 1 m s^{-1}; Figure 2). A formulation that included the dry-snow unloading term (i.e., Equation 12) and its associated coefficient produced a slight reduction in R^2 from 0.32 to 0.2; however, the mean bias and RMSE remained comparable to the formulation based solely on snow load and canopy snowmelt. Based on these results, the relationship combining canopy snow load, snowmelt rate, and shear stress was selected to represent unloading during melt, as it yielded comparable performance and better captures wind-driven unloading:

$$q_{unld}^{melt} = (L \cdot q_{melt}^{veg} \cdot a) + (L \cdot \tau_{mid} \cdot b) \tag{13}$$

where q_{melt}^{veg} is the canopy snowmelt rate (mm hr^{-1}). The comparable performance and similar physics of branch movements for dry and melting snow support the inclusion of this process during melt conditions. Moreover, the larger range of wind speeds observed during dry-snow unloading periods provides a more robust estimate of the shear stress coefficient, which was therefore applied to both non-melt and melt conditions.

Table 3: Regression statistics for melt-induced unloading with various predictor combinations using ordinary least squares (OLS). Error metrics include mean bias (MB), root-mean-squared error (RMSE), coefficient of determination (R^2), and index of agreement (d). Row names (a) and (b) indicate coefficients, with p(a) and p(b) as their p-values. Equation variables include: melt-driven unloading (q_{unld}^{melt}, mm hr^{-1}) and canopy snow melt rate (q_{melt}^{veg}, mm hr^{-1}), with the remaining variables defined in Table 2.

	Snowmelt Rate	Snowmelt Rate, Shear Stress (from non-melt)	Snowmelt Rate, Shear Stress (multiplicative)	Air Temperature	Ice-Bulb Temperature
Fit	OLS	OLS	OLS	OLS	OLS
Eq	$q_{unld}^{melt} = L \cdot q_{melt}^{veg} \cdot a$	$q_{unld}^{melt} = (L \cdot q_{melt}^{veg} \cdot a) + (L \cdot \tau_{mid} \cdot b)$	$q_{unld}^{melt} = L \cdot q_{melt}^{veg} \cdot \tau_{mid} \cdot a$	$q_{unld}^{melt} = L \cdot T_a \cdot a$	$q_{unld}^{melt} = L \cdot T_i \cdot a$
MB	0.161	0.171	0.198	0.128	0.243
RMSE	0.501	0.496	0.477	0.518	0.463
R^2	0.32	0.2	0.26	0.1	0.08
d	0.63	0.59	0.6	0.55	0.44
a	1.05×10^{-01}	9.24×10^{-02}	$6.46 \times 10^{+00}$	1.09×10^{-01}	1.42×10^{-01}
p(a)	$p < 0.05$	$p < 0.05$	$p < 0.05$	$p < 0.05$	$p < 0.05$
b	NA	0.371	NA	NA	NA
p(b)	NA	NA	NA	NA	NA

A limitation of the melt-driven unloading analysis presented in Table 3 is that canopy snow drip and unloading were both measured by the subcanopy weighed snow buckets. To partition melting canopy snow into solid snow unloading and meltwater drip, five warm & humid events were selected, in which the median air temperature was above 0°C and relative humidity was above 65%, resulting in less than 5% contribution of dry-snow unloading and sublimation to ablation as determined by the new model. For these events, unloading was calculated using Equation 11, which combined measurements of total canopy snow ablation from the weighed tree and simulated snowmelt and sublimation from energy-balance based calculations in CRHM. The unloading-to-melt ratio (R, dimensionless) which gives the fraction of canopy snow unloaded due to melt was calculated through a rearrangement of Equation 13 as:

$$R = \frac{q_{unld}^{melt} - (L \cdot \tau_{mid} \cdot b)}{q_{melt}^{veg}} \tag{14}$$

R was found to be positively correlated with canopy snow load (Figure 4). When canopy snow loads ranged from 0 to 5 mm, the unloading-to-melt ratio varied from approximately 0 to 0.5. As snow loads increased, this ratio increased linearly, reaching a peak of 5.0 for a canopy snow load of 30 mm (Figure 4). A relationship for R can be expressed through a linear function:

$$R = m \cdot L + w \tag{15}$$

where m and w are fitting coefficients determined as and , respectively, using the generalised least squares regression shown in Figure 4 (R^2 = 0.73, RMSE = 0.79 (-)). The ordinary least squares regression exhibited heteroscedasticity, with residual variance increasing at higher unloading-to-melt ratios (Breusch-Pagan test $p < 0.05$; see Supporting Information) and was accounted for by using a generalised least squares regression. See the Supporting Information section for additional statistical details.

Additional observations of canopy snowmelt from the subcanopy rain gauges were also used to estimate R (Figure 4). The number of usable observations was limited to three events due to freeze-thaw cycles that seized up the tipping bucket mechanisms in the rain gauges; however, these measurements still provided some validation of the CRHM canopy snowmelt/drip calculations (Figure 5). The CRHM-estimated cumulative drip was higher than the subcanopy rain gauges for two out of the three melt events. Differences in the timing and magnitude of the observed and simulated values were expected due to both instrument uncertainties in the rain gauges from freezing and thawing of snow in the collection funnels, and in the canopy snow energy balance simulation.

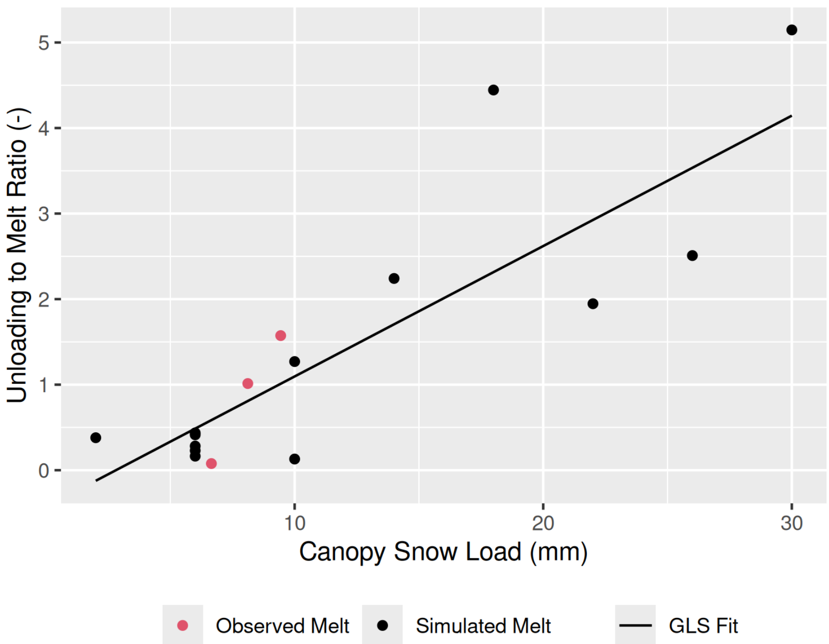

Figure 4: The ratio of canopy snow unloading (weighed tree residual) to snowmelt across different canopy snow load bins for each of the five warm & humid events. Black dots represent the observed cumulative unloading divided by the cumulative snowmelt simulated using the new canopy snow model. Red dots show the cumulative observed unloading divided by snowmelt measured by the rain gauges. Multiple dots within a bin correspond to different events.

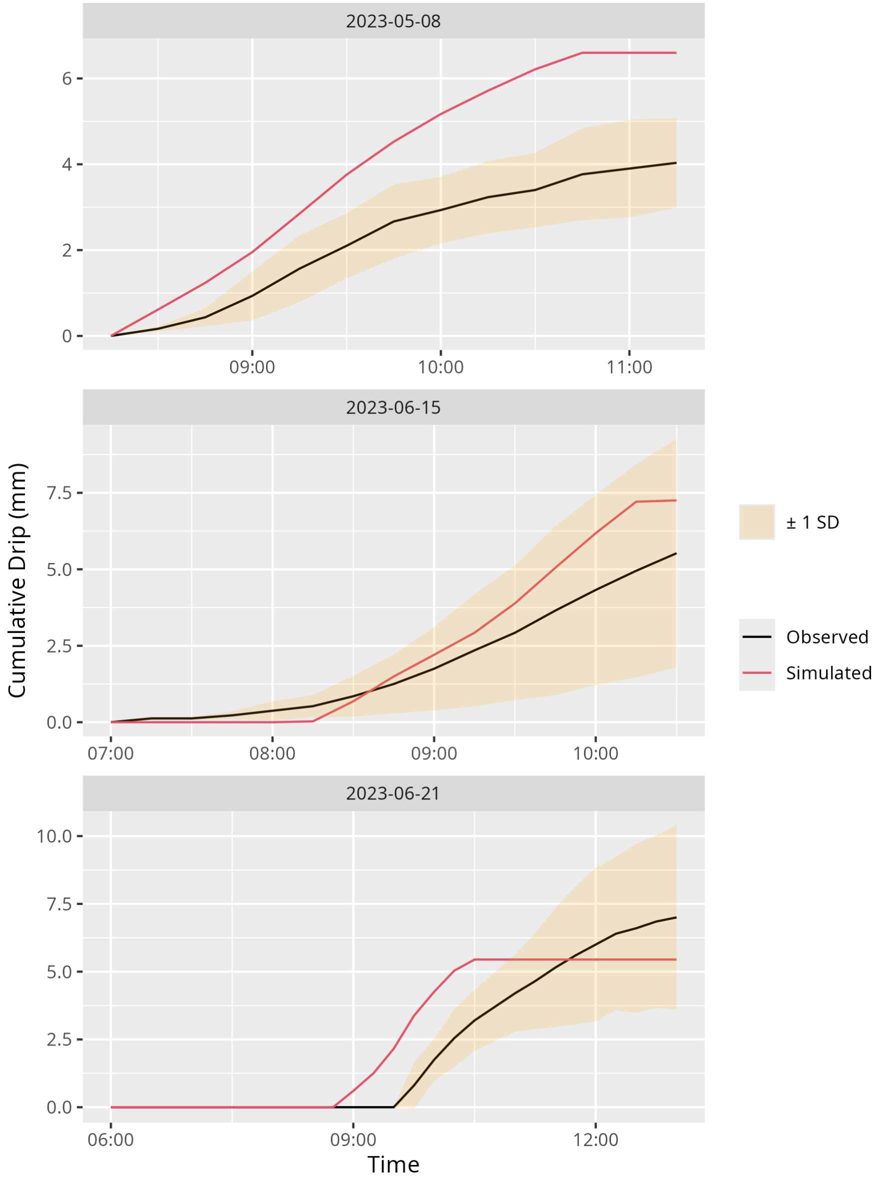

Figure 5: Cumulative canopy snowmelt drip measured as the average of four subcanopy tipping bucket rain gauges (Observed) and simulated using the new canopy snow model for three warm and humid events, with dates labelled as year-month-day.

3.2 New Canopy Snow Model

The new model is based on the canopy snow mass balance formulation (Equation 1), where q_{tf} was represented as:

$$q_{tf} = [(1 - C_p)\alpha]q_{sf} \tag{16}$$

and C_p is the leaf contact area calculated using Equation 10 in Cebulski & Pomeroy (2025b) and α is an efficiency constant.

Since q^{veg}_{melt} can only occur whilst intercepted snow is melting, and wind-induced unloading was found to be representable across all unloading events, the following equation was selected to represent canopy snow unloading q_{unld} (mm hr^{-1}):

$$q_{unld} = (q^{veg}_{melt} \cdot R) + (L \cdot \tau_{mid} \cdot b) \tag{17}$$

where R is calculated from Equation 15, and the coefficient b was selected from the regression analysis over non-melt periods (0.371; see column 2 of Table 3). The remaining terms from Equation 1, such as canopy snow drip (q_{drip}), was derived from calculations of canopy snowmelt from Equation 3, with storage limited by the canopy liquid water holding capacity computed from Equation 2; any excess was assumed to immediately drain. Wind transport of dry canopy snow (q^{veg}_{wind}) was incorporated in the Equation 12 calculation. Sublimation of intercepted snow (q^{sub}_{veg}) was represented using Equation 10.

3.3 Event-based Evaluation of Canopy Snow Ablation Models

The new canopy snow model (referred to as CP25), and the existing models R01, SA09, and E10 (Table 1), were evaluated using weighed tree lysimeter measurements of canopy load during seventeen canopy snow ablation events. These events had air temperatures ranging from -30.5°C to +6.9°C and wind speeds from calm to 5.3 m s^{-1} (Figure 6). This range of meteorological conditions is particularly wide, spanning conditions typical of the cold boreal to temperate maritime needleleaf forests. Events were classified as cold & dry, cold & humid, warm & dry, and warm & humid based on the median event air temperature (above or below 0°C) and relative humidity (above or below 65%) (Figure 6).

Simulated canopy snow load by the new model closely matched the observations for all 17 events, demonstrating the most consistently strong validation amongst the models evaluated (Figure 7). The large decreases in predicted canopy snow load for E10 (Figure 7) are due to the maximum canopy snow load parameter used in this

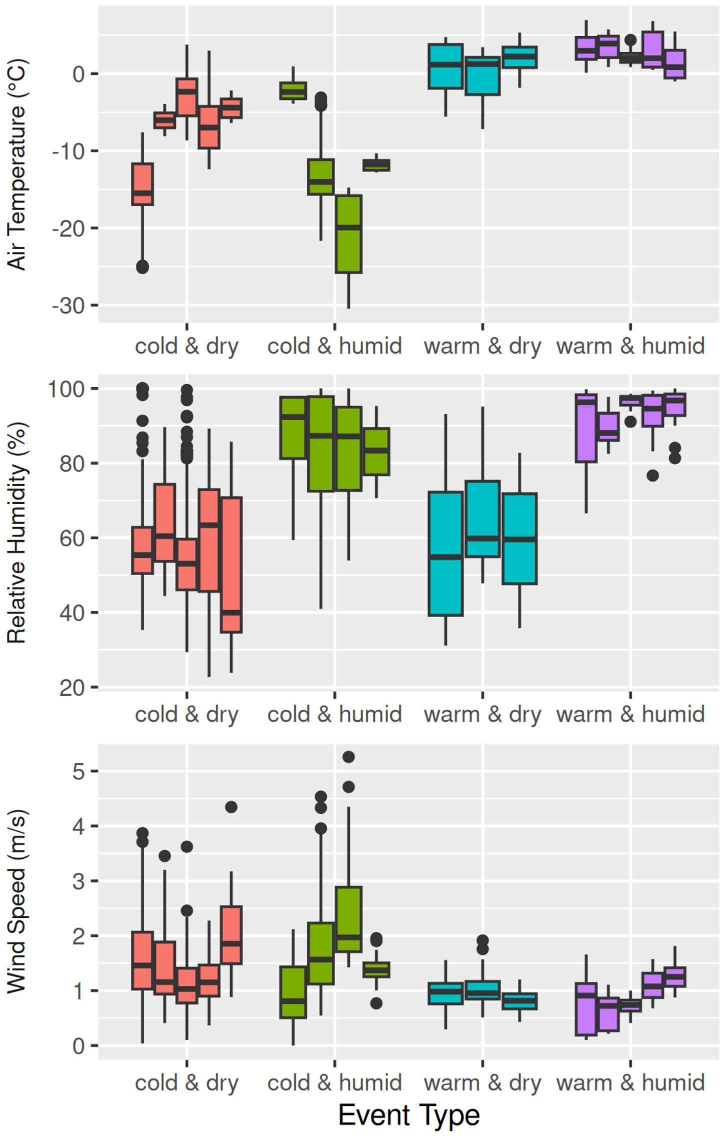

Figure 6: The distribution of observed air temperature, relative humidity, and wind speed over each of the seventeen selected ablation events, grouped into weather types. Note: the rectangle vertical extent represents the interquartile range (25^{th} to 75^{th} percentile), the horizontal line within each box indicates the median, and the whiskers extend to 1.5 times the interquartile range. Circular points beyond the whiskers represent outliers.

model, which ranged from 7 to 12 mm depending on the fresh snow density, which itself is a function of air temperature in (Hedstrom & Pomeroy, 1998). This range in E10 predicted snow loads was much lower than the highest observed snow loads.

The energy balance-based snowmelt modelling approaches (new model & SA09) yielded the most accurate representation of ablation over the warm & humid events. The new model's mean bias of 0.02 mm hr^{-1}, was smaller than the -0.11 mm hr^{-1} bias associated with SA09 (Figure 8). The improvement over SA09 comes from the new model's representation of the increase in unloading at higher canopy snow loads (Figure 4), as observed for the 2022-06-14 (year-month-day) event (Figure 7). The air temperature (R01) and ice-bulb temperature-based (E10) models produced a wider range of mean biases (Figure 8) and larger mean bias of over 0.42 mm hr^{-1} during the warm & humid events, compared to the new model and SA09. The rate of ablation was slower for canopy snow loads below ~1.5 mm and ~0.3 mm for the new model and SA09 respectively; due to their differing liquid water storage capacities and unloading rates (Figure 7). For the warm events other than 2022-04-23, the observed decline in ablation rate occurs around 2 to 3 mm, exceeding the liquid water storage threshold predicted by all models.

The models all had similarly good performance for the warm & dry events, with mean biases of 0.02 mm hr^{-1} for each model (Figure 8). Initiation of ablation was slower than observed from the weighed tree for the new model and SA09 models over all three of the warm & dry events (Figure 7). The temperature threshold methods (E10 & R01) better predicted the timing of the onset of ablation for two events (2022-03-29 and 2022-04-21) compared to the new model. However, E10 predicted an early timing of the onset of ablation and a low ablation rate compared to observations for 2022-04-23.

For the cold & dry events, all models had accurate performance with mean biases ranging from -0.01 to 0.01 mm hr^{-1}, with a slight improvement in the mean bias for the new model of -0.008 mm hr^{-1} (Figure 8). Strong winds approaching 4 m s^{-1} on 2022-03-09 coincided with a rapid decrease in observed snow load, which was poorly captured by all four models, although performance was highest for the new model and R01 which both incorporated wind-driven unloading. For event 2023-03-14, the R01 model overestimated canopy snow ablation due to an overestimation of wind-driven unloading, whereas E10 overpredicted ablation as a result of exceeding the maximum canopy snow storage threshold (Figure 7). However, for E10 this overestimate for one event compensated for underestimates of ablation over the remaining cold & dry

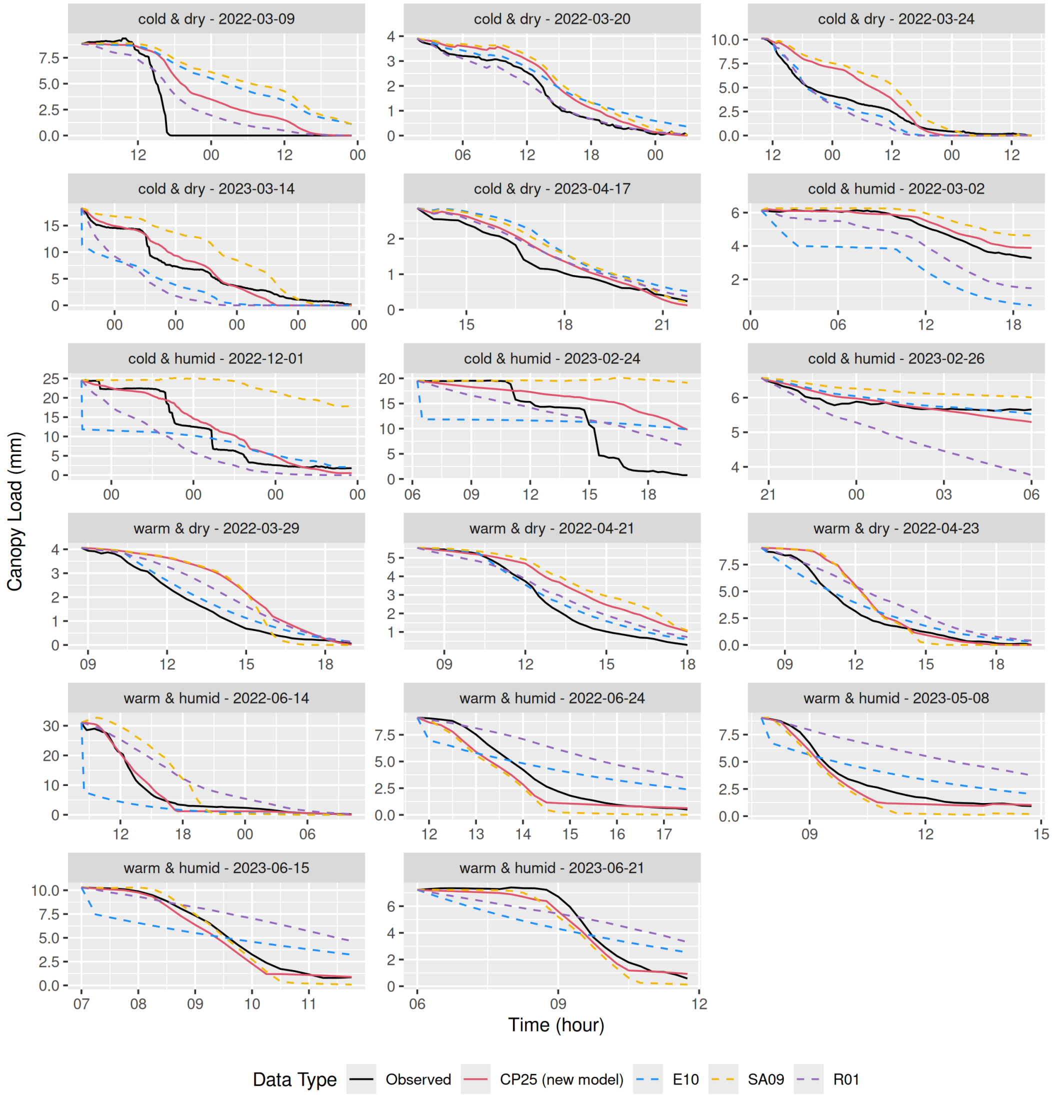

Figure 7: Time series of canopy snow load for individual events measured by the weighed tree (observed) and simulated using the four canopy snow models, with dates labelled as year-month-day. Snow load was simulated using the new model (CP25), Ellis et al. 2010 and Floyd et al. 2012 (E10), Roesch et al. 2001 (R01), and Storck et al. 2002 and Andreadis et al. 2009 (SA09).

events—which had moderate wind speeds—as wind-driven unloading is not included in the E10 model (Figure 6).

The importance of representing wind-driven unloading was clear during the cold & humid events, where the mean bias of models including this process was reduced compared to other approaches; for example, 0.04 mm hr^{-1} for R01 and 0.17 mm hr^{-1} for the new model. In contrast, simulations that did not explicitly account for wind-driven unloading exhibited higher biases, exceeding 0.32 mm hr^{-1} (Figure 8). The R01 model overestimated unloading during most cold & humid events and had higher biases for these events than the new model (Figure 8). For event 2022-03-02, canopy snowmelt was overestimated by the E10 model causing a steep initial decline in canopy snow load which was not captured by the weighed tree or other models. Although the E10 model does not include wind-driven unloading, it performed best for the 2023-02-26 event due to its time-based unloading rate which was slightly slower than the new model and much slower than R01, which greatly overestimated ablation for this event (Figure 7). All models underestimated ablation for the 2023-02-24 event, which had peak wind speeds of over 5 m s^{-1}. Over this event, 1.3 mm of redistributed snowfall was measured under clear sky conditions at a shielded precipitation gauge in a nearby clearing and was likely derived from wind transport of snow from this canopy. The amount of snow observed to unload from the canopy into the subcanopy weighed snow buckets during this event was consistent with simulated unloading in CRHM, suggesting that the remaining unaccounted-for snow was likely entrained into the atmosphere and sublimated and/or transported to distant sites.

3.4 Bootstrap Resampling Evaluation of Canopy Snow Ablation Models

Based on the bootstrap analysis, which evaluated model performance across resampled combinations of canopy snow ablation events, the new model consistently demonstrated the highest performance across all error metrics, resulting in a KGE of 0.63 and lower mean bias and RMSE values (Figure 8). In contrast, the E10, SA09, and R01 models showed the poorest performance, likely due to the absence of key processes leading to KGE values less than 0.41. For example, wind-induced unloading was not represented by the E10 and SA09 models, leading to increased error for wind-driven unloading events (i.e., 2022-03-09; Figure 7). The E10 and R01 models also did not include energy-balance-based melt, and as shown in Figure 8. Consequently, when wind-driven and melt-induced unloading events were included in the bootstrap resampling, errors increased for the E10, SA09, and R01 models relative to the new model, which better represented these processes. However, the inclusion of the maximum canopy snow load parameter in the E10 model provided

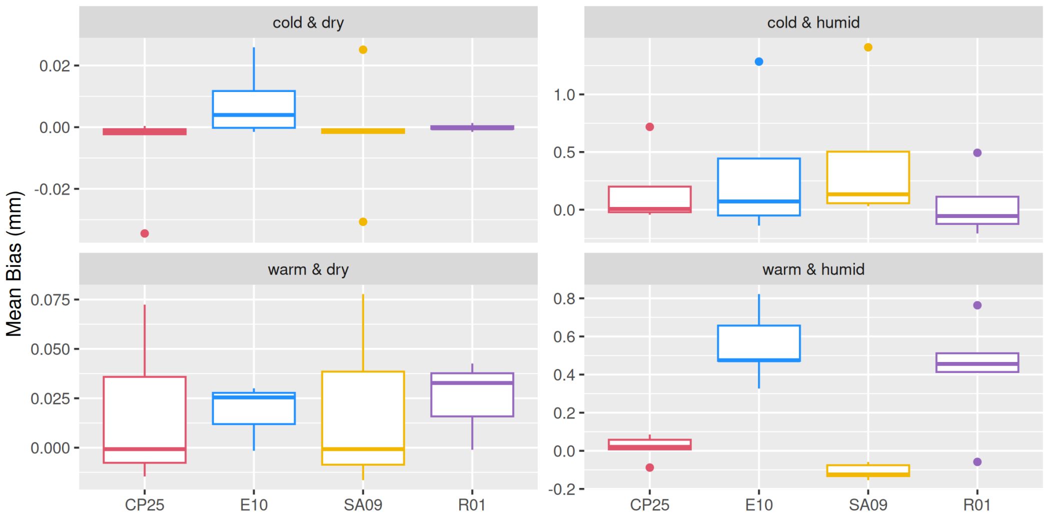

Figure 8: Distribution of mean biases in canopy snow ablation between simulations and observations from the weighed tree for each of the four weather condition groups. Simulated ablation was produced using the new model (CP25), Ellis et al. 2010 and Floyd et al. 2012 (E10), Roesch et al. 2001 (R01), and Storck et al. 2002 and Andreadis et al. 2009 (SA09). The vertical extent of each rectangle represents the interquartile range (25^{th} to 75^{th} percentile), the horizontal line within each box indicates the median, and the whiskers extend to 1.5 times the interquartile range. Circular points beyond the whiskers represent outliers.

some compensation for the underestimation of unloading for larger snow load events (i.e., 2022-12-01; Figure 7). The R01 model exhibited the second-highest performance, which may reflect the inclusion of both melt and wind-induced unloading processes, although its reliance on an air-temperature index melt formulation could explain its reduced accuracy relative to the new model.

3.5 Canopy Snow Partitioning

The partitioning of intercepted snow between evaporative fluxes to the atmosphere via sublimation and evaporation and to the ground via unloading and drip is crucial for assessing the hydrological impact of canopy interception. During warm & humid events, all four parameterisations showed relatively consistent partitioning of canopy snow, with only a small fraction as vapour flux to the atmosphere (Figure 10). The variability of modelled partitioning during warm & dry events was greater than the warm & humid events, with a larger contribution from sublimation and evaporation for warm & dry events. The E10 model predicted a greater fraction of intercepted snow unloading to the ground than did other models for the warm & dry events.

For the cold & dry and cold & humid events, the two parameterisations that include wind-driven unloading (R01 and the new model) estimated differing fractions of snow

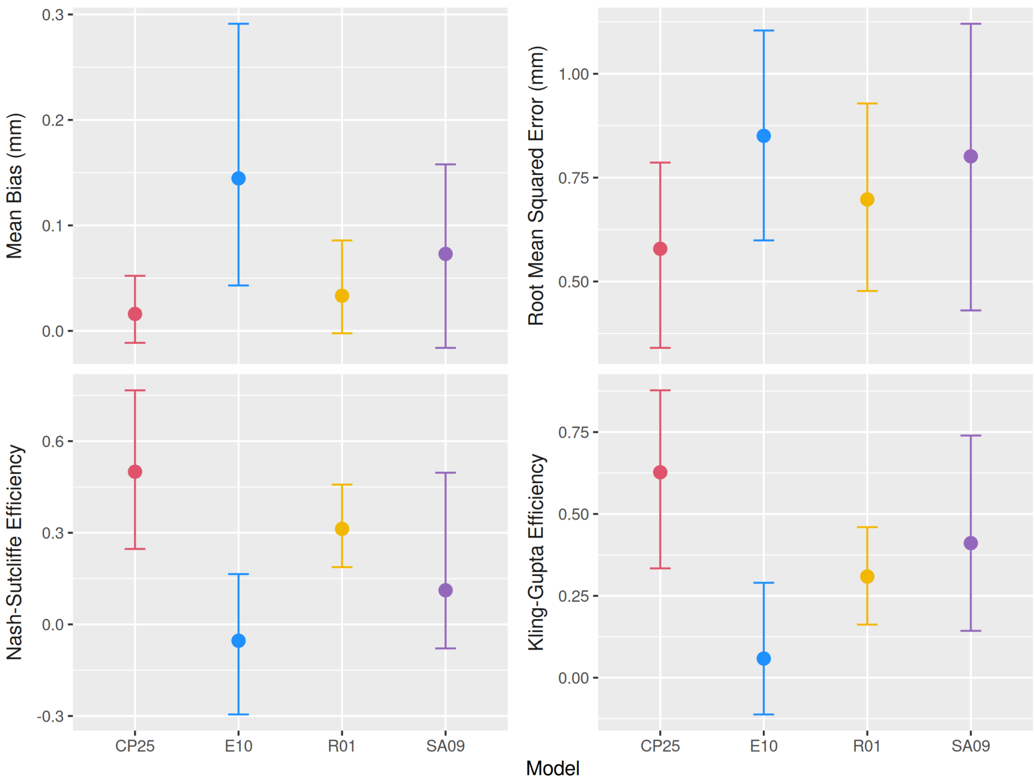

Figure 9: Error statistics derived from bootstrap resampling of differing combinations of canopy snow ablation events (5000 replicates). Simulated ablation was produced using the new model (CP25), Ellis et al. 2010 and Floyd et al. 2012 (E10), Roesch et al. 2001 (R01), and Storck et al. 2002 and Andreadis et al. 2009 (SA09). Points indicate the mean metric and error bars show the 95% confidence intervals estimated across all resampled event combinations.

unloading, where R01 simulated a higher fraction of intercepted snow unloading. For both the cold & dry and cold & humid events, SA09 partitioned all snow back to the atmosphere via sublimation, as dry-snow unloading is not included in this parameterisation. Although the new model and E10 differ in their unloading processes, when averaged across all events, both showed a similar fraction of snow reaching the ground (70%) versus the atmosphere (30%). The difference in intercepted snowfall partitioning across all events was larger for the SA09 and R01 models, which had 40% and 24% of snow reaching the atmosphere respectively.

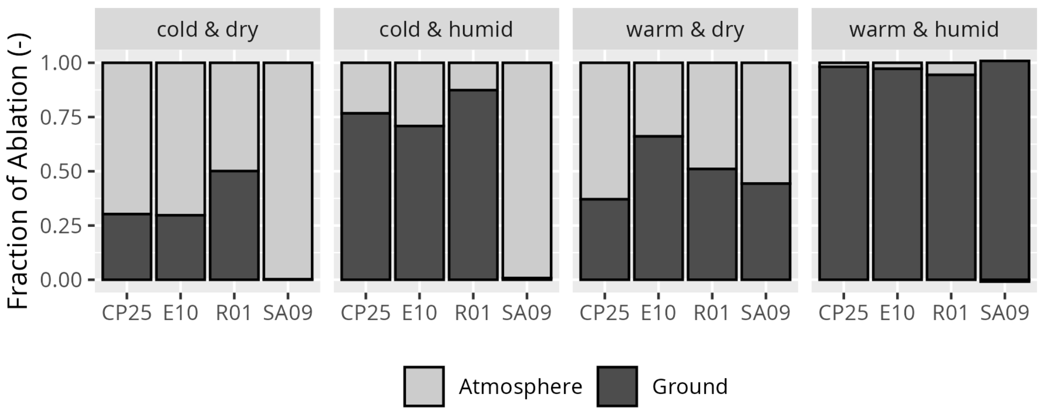

Figure 10: The proportion of intercepted snow that was either lost to the atmosphere as sublimation and/or evaporation or transferred to the ground through unloading or drip of melted snow by each weather type for all 17 events. Simulated ablation was produced using the new model (CP25), Ellis et al. 2010 and Floyd et al. 2012 (E10), Roesch et al. 2001 (R01), and Storck et al. 2002 and Andreadis et al. 2009 (SA09).

4 Discussion

4.1 Processes Governing Canopy Snow Unloading

Observations of canopy snow unloading support the hypothesis that unloading is primarily controlled by canopy snow load, wind shear stress, and canopy snow melt. These results differ from earlier research that represented wind-driven unloading as a linear function of wind speed and canopy load (Bartlett & Verseghy, 2015; Katsushima et al., 2023; Roesch et al., 2001). The slightly higher performance of the shear stress relationship compared to wind speed—when excluding melt events—is likely due to the physical relationship between shear stress force and kinetic energy transfer to the canopy, wind transport from the canopy, and movement of branches in the canopy induced by drag (shear) forces. The slightly lower wind-driven unloading rate presented in this study, compared to the R01 model, may be attributed to the development of the R01 parameterisation using above canopy albedo as a proxy for canopy snow unloading (Bartlett & Verseghy, 2015; Roesch et al., 2001). This approach would have included both unloading and sublimation processes, in addition to greater measurement uncertainties (Cebulski & Pomeroy, 2025a), and may have contributed to R01 overestimating wind-driven unloading (Figure 8). Conversely, the measurements collected here provided a more direct quantification of canopy snow unloading rates. Additional factors not considered in the new parameterisations include structural degradation and bond weakening from sublimation, snow density, and branch elasticity. These factors could all directly affect unloading relationships. For instance, variation in branch flexibility amongst tree species and age classes could alter how snow load and wind affects branch angle and swaying, thereby changing the unloading relationships. Sublimation of intercepted snow reduces its structural integrity (MacDonald, 2010), whereas equi-temperature metamorphism can increase density and cohesion (Pomeroy & Gray, 1995). Both processes affect canopy snow unloading rates and the exposure of intercepted snow to turbulent energy exchanges; however, neither process emerged in this study. Fresh, low-density snow typically exhibits lower cohesion and adhesion than older snow that has undergone freeze-thaw cycles or equi-temperature metamorphism (Pomeroy & Gray, 1995).

The ratio of unloading to canopy snowmelt increased linearly with increasing canopy snow load (Figure 4), unlike Storck et al. (2002), who originally found the ratio to be constant. The measurement difficulties noted by Storck et al. (2002) limited their estimate of this ratio to a single event, preventing any association with canopy snow load. Previous studies have identified relationships between melt-induced unloading and various meteorological parameters, including empirical functions of air temperature (Katsushima et al., 2023; Roesch et al., 2001), ice-bulb temperature (Ellis et al., 2010; Floyd, 2012), and solar radiation (Katsushima et al., 2023). Although branch bending and subsequent unloading have been shown to be associated with air temperature (Schmidt & Gluns, 1991; Schmidt & Pomeroy, 1990), observations here indicate that temperature-related unloading increases occur primarily near the freezing point, eliminating the need for a separate temperature parameterisation beyond snowmelt-associated unloading processes. The implementation of wind-driven unloading when canopy snow is melting was supported by the similar performance with and without including a snow load and shear stress term (Table 3). Moreover, the similar physics of branch swaying under both canopy snowmelt and non-melt conditions provides justification for using the same wind-driven unloading relationship for melting and non-melting snow. This also allows for a single continuous function to be used to calculate canopy snow unloading over both melt and non-melt conditions (Equation 17). Calm winds were observed over the canopy snowmelt periods and limited the testing of including shear stress as a predictor of unloading over these conditions in this study. The addition of liquid water content due to phase change of canopy snow can increase resistance to wind transport (Pomeroy & Gray, 1995) or decrease cohesion and adhesion (Storck et al. (2002); Figure 4). Thus, further research is required to evaluate the wind-driven unloading process over a greater range of wind speeds while canopy snow is melting.

4.2 Comparison of Ablation Model Performance

The improved performance of the new model across a wide range of meteorological conditions demonstrates the advantage of incorporating snow load, shear stress, and melt-driven unloading processes coupled with a physically based energy balance to simulate ablation (Figure 9). In contrast, existing models were limited by missing processes such as wind-driven unloading (SA09 and E10) or relied on temperature-dependent parameterisations of melt and drip processes, which had limited transferability across differing events (R01 and E10). The constant canopy snow unloading to melt ratio of 0.4 in the SA09 model led to reduced performance for one melt event with canopy snow loads up to 30 mm and for all melt events when snow loads were less than 1 mm, whereas the new model more accurately predicted ablation for a range of snow loads during the melt events (Figure 7). The lower liquid water retention capacity and increased solid snow unloading for lower snow loads contributed to SA09's over estimation of ablation during the end of most melt events (Figure 7).

Models using empirical temperature-based canopy snowmelt parameterisations (E10 and R01) exhibited inconsistent performance, particularly during warm & humid events where they underestimated intercepted snow ablation (Figure 8). This was due to their reliance on air or ice-bulb temperature as proxies for energy input into the canopy, which failed to accurately represent available energy during these events. In contrast, all four models performed similarly during warm & dry events, when air temperatures were closer to 0°C, and lower humidity increased the partitioning of available energy into canopy snow sublimation. The superior performance of the energy balance-based canopy snowmelt models (the new model and SA09) during warm & humid events likely reflects their ability to accurately represent the energy in these conditions. The maximum canopy snow load threshold implemented in the E10 model was lower than observations from the weighed tree. This limitation offset its tendency to underestimate unloading processes, as observed in Figure 7 and also by Lundquist et al. (2021) and Lumbrazo et al. (2022). In this study, the new model and the SA09 and R01 models did not include a maximum snow load, and their performance aligns with the hypothesis of Lundquist et al. (2021) that this limit may be unnecessary—or is much higher than previously thought—when combined with a comprehensive canopy snow ablation routine.

Although the new model provided a better representation of wind-driven unloading than R01 (Figure 8); substantial errors persisted across all models during windy events. Wind unloading parameters may also be influenced by tree species, forest structure, and snow structural characteristics (Lumbrazo et al., 2022), necessitating further research on canopy snow unloading across diverse environments to assess their broader applicability. All models underestimated ablation for the strongest wind-dominated event (2023-02-24, Figure 7). Wind transport of canopy snow to a nearby gap occurred during this event, but was a small fraction of canopy snow ablation, 1.3 mm of a total of 20 mm of canopy snow ablation. Since unloading measured by the subcanopy weighed snow buckets corresponded well with simulations from the new model for this event, the approximately 9 mm of unaccounted canopy snow ablation may be attributed to uncertainties within the wind-driven unloading parameterisation (Figure 3) and possible atmospheric entrainment of canopy snow that was either transported to distant locations and/or sublimated. These findings align with observations by Troendle (1983) but contrast with Hoover & Leaf (1967), who proposed that most wind-transported snow relocates to nearby sites with minimal influence of sublimation.

4.3 Canopy Snow Partitioning

Substantial variability amongst snow ablation model predictions was found in the fraction of snow that sublimated and/or meltwater that evaporated to the atmosphere versus unloading and drip to the ground (Figure 10). For example, the lack of cold snow unloading processes in the SA09 model resulted in 100% of intercepted snow reaching the atmosphere via canopy snow sublimation for both the cold & dry and cold & humid events. This differed considerably from the new model, which returned 70% and 24% of snow back to the atmosphere for the cold & dry and cold & humid events, respectively. Although E10 and the new model differ in their process representations, they predict comparable fractions of snow reaching the ground versus returning to the atmosphere (Figure 10). The agreement between the new model and E10 is notable, as E10 has been tested and found to perform well at predicting subcanopy snowpacks worldwide (Ellis et al., 2010; Gelfan et al., 2004; Pomeroy et al., 2022; Rutter et al., 2009; Sanmiguel-Vallelado et al., 2022). However, the results shown here reveal that E10's individual process representations can be in error, particularly under warm, windy conditions, potentially explaining difficulties when applying E10 at locations where parameterisation errors fail to compensate for one another. For example, canopies that intercept a larger amount of snow, the E10 maximum canopy snow load tends to overestimate the amount of unloading as shown in this study and in Lundquist et al. (2021) and Lumbrazo et al. (2022).

4.4 Future Directions

Physically based approaches are particularly relevant for predicting snow regimes in forested basins under a changing climate, where warming may reduce the reliability of empirically derived canopy snowmelt models such as E10 and R01. Many regions are shifting, or are expected to shift from cold dry environments, where snow regimes are dominated by canopy snow unloading and sublimation, to warmer and wetter environments with increased drip and melt-driven unloading processes (Elshamy et al., 2025; e.g., Fang & Pomeroy, 2020). Although the new model performed best for this dataset, further validation is required across a wider range of climates and forest structures.

Lumbrazo et al. (2022) suggest that the unloading rate of rime-ice is very low. Vapour deposition and rime-ice accumulation are simulated in some models (e.g., Clark et al., 2015; Ellis et al., 2010) and in the new model via sublimation parameterisation, as additions to the canopy snow load. However, in the humid, temperate springtime observations presented in this study, canopy riming and frost deposition were not detected. Thus, the snow unloading rates presented here could overestimate unloading rates for ice intercepted in the canopy. This introduces uncertainty in the transferability of these results to maritime climates such as the Coastal Mountains of Canada, where rime and frost deposition occur more frequently. The addition of a canopy ice reservoir to hold rime-ice, frost, and ice from melt-freeze cycles might help better represent differing ablation process and should be studied in maritime environments.

This study implemented a single-layer snow model for snow intercepted in the canopy. The unloading relationships developed here, along with the parameterisation for initial interception presented in Cebulski & Pomeroy (2025b), are expected to be applicable to multi-layer canopy snow models (e.g., Clark et al., 2015; Essery et al., 2025), with modifications to the energy balance calculations described in Section 2.5. However, uncertainties in generating forcing data and collecting canopy snow ablation measurements at different canopy heights may impede the implementation, performance, and assessment of these vertically distributed relationships.

Uncertainties remain in measuring canopy snow sublimation using eddy correlation systems (Conway et al., 2018; Harding & Pomeroy, 1996; Harvey et al., 2025;

Helgason & Pomeroy, 2012b; Parviainen & Pomeroy, 2000) and in separating snow unloading from meltwater drip (Floyd, 2012; Storck et al., 2002), which limit the development and testing of canopy snow ablation parameterisations. Since unloading, melt, and sublimation are competitive ablation processes, correctly representing each of them is essential for determining the fraction and phase of snow reaching the ground, and the duration that the canopy is snow covered, an important factor for land-surface albedo calculations (Cai et al., 2025; Henderson et al., 2018; Thackeray et al., 2014). Whilst separating initial interception and ablation processes (Cebulski & Pomeroy, 2025a, 2025b) improves process representations, these routines still need to be evaluated together against additional field observations across different sites, climates, forest types, and spatial scales to assess model transferability and performance.

5 Conclusions

Canopy snow ablation processes govern how intercepted snowfall is partitioned to either the ground or the atmosphere, yet their representation in models remained uncertain due to insufficient validation. This study represents the first development and process-level validation of an unloading model addressing both dry-snow and melt-driven unloading coupled to energy-balance based calculations of canopy snowmelt and sublimation. Canopy snow load, shear stress, and canopy snowmelt were found to be the best predictors of unloading, with key differences from previously established approaches. A new finding was that wind-driven unloading was strongly related to snow load and shear stress, and that the ratio of unloading to canopy snowmelt increased with canopy snow load. The new model performed better than existing models that focused exclusively on snow unloading from melt or used temperature index melt routines. In addition to this empirical evidence, consideration of likely physical processes such as intercepted snow clump structural degradation due to bond weakening, lubrication of wet snow and branches during melt, and the entrainment of snow and branch swaying due to shear forces exerted on the canopy by wind further support representing these processes. Moreover, an existing approach that used the concept of a maximum initial intercepted snow load greatly underestimated canopy snow storage capacity compared with snow loads observed from weighed tree measurements. Throughout the two years of observations presented here, a maximum initial canopy snow load was not observed, likely as unloading rates increased with higher snow loads. Wind transport events were relatively rare even in this wind-exposed subalpine forest but resulted in a considerable underestimation of the amount of snow ablated to the atmosphere or redistributed surrounding sites during one particularly windy event.

A new canopy snow ablation model that incorporates an updated canopy snow mass and energy balance demonstrated improved accuracy in simulating canopy snow ablation compared to existing approaches under a wide range of meteorological conditions. Existing models failed to maintain accuracy across these conditions due to neglect of key processes such as wind-driven unloading, dry snow unloading, and/or energy balance-based representations of melt processes. The greatest inter-model discrepancies in canopy snow load occurred during warm, humid weather conditions, during which temperature-based canopy snowmelt parameterisations showed substantially higher errors than energy balance-based models.

Amongst the models tested, the largest errors occurred during cold, humid, and windy unloading events, though performance improved when incorporating a shear stress-based unloading parameterisation. Model partitioning of intercepted snow between the ground and atmosphere differed most during cold weather events, where neglected dry-snow unloading processes greatly overestimated canopy snow sublimation. Observations at this sub-alpine site were collected over a wide range of weather conditions that characterised typical weather ranging from cold boreal to temperate maritime climates, with air temperatures from -30.5°C to +6.7°C and wind speeds from calm to 5.3 m s^{-1}. Some processes, such as rime-ice and frost deposition, were not observed. The study was conducted in a sparse forest of mature fir and spruce, so unloading rates and process emergence may differ in forests with other species, densities, and ages due to differences in branch elasticity, structure, and associated impacts on the canopy snow mass and energy balance. Incorporating the updated unloading schemes developed here could improve the representation of canopy snow ablation and, by extension, the partitioning of precipitation between surface accumulation and vapour exchange with the atmosphere and through more accurate snow load estimation and thus canopy albedo in hydrological and land surface models. Nonetheless, further testing is needed across various climate and forest compositions to assess model transferability.

6 Acknowledgements

We acknowledge financial support from the University of Saskatchewan Dean's Scholarship, the Natural Sciences and Engineering Research Council of Canada's Discovery Grants, the Canada First Research Excellence Fund's Global Water Futures Programme, Environment and Climate Change Canada, Alberta Innovates Water Innovation Program, the Canada Foundation for Innovation's Global Water Futures

Observatories facility, and the Canada Research Chairs Programme. We thank Madison Harasyn, Hannah Koslowsky, Kieran Lehan, Lindsey Langs, and Fortress Mountain Resort for their help in the field, and Tom Brown and Logan Fang for their support of the CRHM platform. The authors thank Dr. Cherie Westbrook, Dr. Martyn Clark, and Dr. Warren Helgason for their valuable feedback on this manuscript.

7 Data & Software Availability Statement

The Cold Regions Hydrological Model Platform (CRHM) source code is available at https://github.com/srlabUsask/crhmcode. Model forcing data, model outputs, validation data, processed data, and scripts are available at https://doi.org/10.5281/zenodo.16898881.

References

Allan, R., Pereira, L., Raes, D., & Smith, M. (1998). *Crop Evapotranspiration Guidelines for Computing Crop Water Requirements*. Food and Agriculture Organization of the United Nations.

Andreadis, K. M., Storck, P., & Lettenmaier, D. P. (2009). Modeling snow accumulation and ablation processes in forested environments. *Water Resources Research*, *45*(5), 1–33. https://doi.org/10.1029/2008WR007042

Aubry-Wake, C., & Pomeroy, J. W. (2023). Predicting hydrological change in an alpine glacierized basin and its sensitivity to landscape evolution and meteorological forcings. *Water Resources Research*, *59*(9). https://doi.org/10.1029/2022WR033363

Baggi, S., & Schweizer, J. (2008). Characteristics of wet-snow avalanche activity: 20 years of observations from a high alpine valley (Dischma, Switzerland). *Natural Hazards*, *50*(1), 97–108. https://doi.org/10.1007/s11069-008-9322-7

Bartlett, P. A., & Verseghy, D. L. (2015). Modified treatment of intercepted snow improves the simulated forest albedo in the Canadian Land Surface Scheme. *Hydrological Processes*, *29*(14), 3208–3226. https://doi.org/10.1002/HYP.10431

Cai, L., Ma, J., Zhang, Y., Shi, W., & Yang, R. (2025). Boreal forest cover change since 2000 contributes to cold winters in Eurasia. *Npj Climate and Atmospheric Science*, *8*(1), 337. https://doi.org/10.1038/s41612-025-01219-1

Cebulski, A. C., & Pomeroy, J. W. (2025a). Theoretical underpinnings of snow interception and canopy snow ablation parameterisations. *WIREs Water*, *12*, e70010. https://doi.org/10.1002/wat2.70010

Cebulski, A. C., & Pomeroy, J. W. (2025b). Snow interception relationships with meteorology and canopy density. *Hydrological Processes*, *39*(4), e70135. https://doi.org/10.1002/hyp.70135

Clark, M. P., Nijssen, B., Lundquist, J. D., Kavetski, D., Rupp, D. E., Woods, R. A., Freer, J. E., Gutmann, E. D., Wood, A. W., Gochis, D. J., Rasmussen, R. M., Tarboton, D. G., Mahat, V., Flerchinger, G. N., & Marks, D. G. (2015). A unified approach for process-based hydrologic modeling: 2. Model implementation and case studies. *Water Resources Research*, *51*(4), 2515–2542. https://doi.org/10.1002/2015WR017200

Clark, M. P., Vogel, R. M., Lamontagne, J. R., Mizukami, N., Knoben, W. J. M., Tang, G., Gharari, S., Freer, J. E., Whitfield, P. H., Shook, K. R., & Papalexiou, S. M. (2021). The abuse of popular performance metrics in hydrologic modeling. *Water Resources Research*, *57*(9), 1–16. https://doi.org/10.1029/2020WR029001

Conway, J. P., Pomeroy, J. W., Helgason, W. D., & Kinar, N. J. (2018). Challenges in modeling turbulent heat fluxes to snowpacks in forest clearings. *Journal of Hydrometeorology*, *19*(10), 1599–1616. https://doi.org/10.1175/JHM-D-18-0050.1

Deschamps-Berger, C., López-Moreno, J. I., Gascoin, S., Mazzotti, G., & Boone, A. (2025). Where snow and forest meet: A global atlas. *Geophysical Research Letters*, *52*(10), e2024GL113684. https://doi.org/10.1029/2024GL113684

Ellis, C. R., Pomeroy, J. W., Brown, T., & MacDonald, J. (2010). Simulation of snow accumulation and melt in needleleaf forest environments. *Hydrology and Earth System Sciences*, *14*(6), 925–940. https://doi.org/10.5194/hess-14-925-2010

Elshamy, M. E., Pomeroy, J. W., Pietroniro, A., Wheater, H., & Abdelhamed, M. (2025). The Impact of Climate and Land Cover Change on the Cryosphere and Hydrology of the Mackenzie River Basin, Canada. *Water Resources Research*, *61*(12), e2024WR039276. https://doi.org/10.1029/2024WR039276

Essery, R., Mazzotti, G., Barr, S., Jonas, T., Quaife, T., & Rutter, N. (2025). A Flexible Snow Model (FSM 2.1.0) including a forest canopy. *EGUsphere*, 1–37. https://doi.org/10.5194/egusphere-2024-2546

Essery, R., Pomeroy, J. W., Parviainen, J., & Storck, P. (2003). Sublimation of snow from coniferous forests in a climate model. *Journal of Climate*, *16*(11), 1855–1864. https://doi.org/10.1175/1520-0442(2003)016%3C1855:SOSFCF%3E2.0.CO;2

Fang, X., & Pomeroy, J. W. (2020). Diagnosis of future changes in hydrology for a Canadian Rocky Mountain headwater basin. *Hydrology and Earth System Sciences Discussions*, 1–40. https://doi.org/10.5194/hess-2019-640

Fang, X., & Pomeroy, J. W. (2023). Simulation of the impact of future changes in climate on the hydrology of Bow River headwater basins in the Canadian Rockies. *Journal of Hydrology*, *620*, 129566. https://doi.org/10.1016/j.jhydrol.2023.129566

Floyd, W. C. (2012). *Snowmelt Energy Flux Recovery During Rain-on-Snow in Regenerating Forests* (p. 180) [PhD thesis, University of British Columbia]. https://doi.org/10.14288/1.0073024

Gelfan, A. N., Pomeroy, J. W., & Kuchment, L. S. (2004). Modeling forest cover influences on snow accumulation, sublimation, and melt. *Journal of Hydrometeorology*, *5*(5), 785–803. https://doi.org/10.1175/1525-7541(2004)005%3C0785:MFCIOS%3E2.0.CO;2

Gupta, H. V., Kling, H., Yilmaz, K. K., & Martinez, G. F. (2009). Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *Journal of Hydrology*, *377*(1-2), 80–91. https://doi.org/10.1016/j.jhydrol.2009.08.003

Harder, P., & Pomeroy, J. W. (2013). Estimating precipitation phase using a psychrometric energy balance method. *Hydrological Processes*, *27*(13), 1901–1914. https://doi.org/10.1002/hyp.9799

Harding, R. J., & Pomeroy, J. W. (1996). The energy balance of the winter boreal landscape. *Journal of Climate*, *9*(11), 2778–2787. https://www.jstor.org/stable/26201420

Harvey, N., Burns, S. P., Musselman, K. N., Barnard, H., & Blanken, P. D. (2025). Identifying canopy snow in subalpine forests: A comparative study of methods. *Water Resources Research*, *61*(1), e2023WR036996. https://doi.org/10.1029/2023WR036996

Hedstrom, N. R., & Pomeroy, J. W. (1998). Measurements and modelling of snow interception in the boreal forest. *Hydrological Processes*, *12*(10-11), 1611–1625.

Helgason, W., & Pomeroy, J. W. (2012a). Characteristics of the near-surface boundary layer within a mountain valley during winter. *Journal of Applied Meteorology and Climatology*, *51*(3), 583–597. https://doi.org/10.1175/JAMC-D-11-058.1

Helgason, W., & Pomeroy, J. W. (2012b). Problems closing the energy balance over a homogeneous snow cover during midwinter. *Journal of Hydrometeorology*, *13*(2), 557–572. https://doi.org/10.1175/JHM-D-11-0135.1

Henderson, G. R., Peings, Y., Furtado, J. C., & Kushner, P. J. (2018). Snow–atmosphere coupling in the northern hemisphere. *Nature Climate Change*, *8*(11), 954–963. https://doi.org/10.1038/s41558-018-0295-6

Hoover, M. D., & Leaf, C. F. (1967). Processs and Significance of Interception in Colorado Subalpine Forest. *Proceeding of a National Science Foundation Advanced Science Seminar.*

Immerzeel, W. W., Lutz, A. F., Andrade, M., Bahl, A., Biemans, H., Bolch, T., Hyde, S., Brumby, S., Davies, B. J., Elmore, A. C., Emmer, A., Feng, M., Fernández, A., Haritashya, U., Kargel, J. S., Koppes, M., Kraaijenbrink, P. D. A., Kulkarni, A. V., Mayewski, P. A., … Baillie, J. E. M. (2020). Importance and vulnerability of the world's water towers. *Nature*, *577*(7790), 364–369. https://doi.org/10.1038/s41586-019-1822-y

Katsushima, T., Kato, A., Aiura, H., Nanko, K., Suzuki, S., Takeuchi, Y., & Murakami, S. (2023). Modelling of snow interception on a Japanese cedar canopy based on weighing tree experiment in a warm winter region. *Hydrological Processes*, *37*(6), 1–16. https://doi.org/10.1002/hyp.14922

Kesselring, J., Morsdorf, F., Kükenbrink, D., Gastellu-Etchegorry, J.-P., & Damm, A. (2024). Diversity of 3D APAR and LAI dynamics in broadleaf and coniferous forests: Implications for the interpretation of remote sensing-based products. *Remote Sensing of Environment*, *306*, 114116. https://doi.org/10.1016/j.rse.2024.114116

Kim, E., Gatebe, C., Hall, D., Newlin, J., Misakonis, A., Elder, K., Marshall, H. P., Hiemstra, C., Brucker, L., De Marco, E., Crawford, C., Kang, D. H., & Entin, J. (2017). NASA's snowex campaign: Observing seasonal snow in a forested environment. *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, 1388–1390. https://doi.org/10.1109/IGARSS.2017.8127222

Kozak, A., & Kozak, R. A. (1995). Notes on regression through the origin. *Forestry Chronicle*, *71*(3), 326–330. https://doi.org/10.5558/tfc71326-3

Krinner, G., Derksen, C., Essery, R., Flanner, M., Hagemann, S., Clark, M. P., Hall, A., Rott, H., Brutel-Vuilmet, C., Kim, H., Ménard, C. B., Mudryk, L., Thackeray, C., Wang, L., Arduini, G., Balsamo, G., Bartlett, P., Boike, J., Boone, A., … Zhu, D. (2018). ESM-SnowMIP: Assessing snow models and quantifying snow-related climate feedbacks. *Geoscientific Model Development*, *11*(12), 5027–5049. https://doi.org/10.5194/gmd-11-5027-2018

López-Moreno, J. I., Zabalza, J., Vicente-Serrano, S. M., Revuelto, J., Gilaberte, M., Azorin-Molina, C., Morán-Tejeda, E., García-Ruiz, J. M., & Tague, C. (2014). Impact of climate and land use change on water availability and reservoir management: Scenarios in the Upper Aragón River, Spanish Pyrenees. *Science of The Total Environment*, *493*, 1222–1231. https://doi.org/10.1016/j.scitotenv.2013.09.031

Lumbrazo, C., Bennett, A., Currier, W. R., Nijssen, B., & Lundquist, J. (2022). Evaluating multiple canopy-snow unloading parameterizations in SUMMA with time-lapse photography characterized by citizen scientists. *Water Resources Research*, *58*(6), 1–22. https://doi.org/10.1029/2021WR030852

Lundquist, J. D., Dickerson-Lange, S., Gutmann, E., Jonas, T., Lumbrazo, C., & Reynolds, D. (2021). Snow interception modelling: Isolated observations have led to many land surface models lacking appropriate temperature sensitivities. *Hydrological Processes*, *35*(7), 1–20. https://doi.org/10.1002/hyp.14274

MacDonald, J. P. (2010). *Unloading of Intercepted Snow in Conifer Forests* (p. 93) [Msc Thesis]. University of Saskatchewan.

Mazzotti, G., Webster, C., Essery, R., & Jonas, T. (2021). Increasing the physical representation of forest-snow processes in coarse-resolution models: Lessons learned from upscaling hyper-resolution simulations. *Water Resources Research*, *57*(5), 1–21. https://doi.org/10.1029/2020WR029064

Nash, J. E., & Sutcliffe, J. V. (1970). River flow forecasting through conceptual models part I — A discussion of principles. *Journal of Hydrology*, *10*(3), 282–290. https://doi.org/10.1016/0022-1694(70)90255-6

Parviainen, J., & Pomeroy, J. W. (2000). Multiple-scale modelling of forest snow sublimation: Initial findings. *Hydrological Processes*, *14*(15), 2669–2681. https://doi.org/10.1002/1099-1085(20001030)14:15%3C2669::AID-HYP85%3E3.0.CO;2-Q

Pomeroy, J. W., Brown, T., Fang, X., Shook, K. R., Pradhananga, D., Armstrong, R., Harder, P., Marsh, C., Costa, D., Krogh, S. A., Aubry-Wake, C., Annand, H., Lawford, P., He, Z., Kompanizare, M., & Moreno, J. I. L. (2022). The cold regions hydrological modelling platform for hydrological diagnosis and prediction based on process understanding. *Journal of Hydrology*, *615*(128711), 1–25. https://doi.org/10.1016/j.jhydrol.2022.128711

Pomeroy, J. W., & Gray, D. M. (1995). *Snowcover Accumulation, Relocation and Management* (NHRI Science Report No. 7, p. 144). National Hydrology Research Institute, Environment Canada, Saskatoon, Canada.

Pomeroy, J. W., Gray, D. M., Hedstrom, N. R., & Janowicz, J. R. (2002). Prediction of seasonal snow accumulation in cold climate forests. *Hydrological Processes*, *16*(18), 3543–3558. https://doi.org/10.1002/hyp.1228

Pomeroy, J. W., Gray, D. M., Shook, K. R., Toth, B., Essery, R. L. H. H., Pietroniro, A., & Hedstrom, N. (1998a). An evaluation of snow accumulation and ablation processes for land surface modelling. *Hydrological Processes*, *12*(15), 2339–2367. https://doi.org/10.1002/(SICI)1099-1085(199812)12:15%3C2339::AID-HYP800%3E3.0.CO;2-L

Pomeroy, J. W., Marks, D., Link, T., Ellis, C. R., Hardy, J., Rowlands, A., & Granger, R. (2009). The impact of coniferous forest temperature on incoming longwave radiation to melting snow. *Hydrological Processes*, *23*, 2513–2525. https://doi.org/10.1002/hyp.7325

Pomeroy, J. W., Parviainen, J., Hedstrom, N., & Gray, D. M. (1998b). Coupled modelling of forest snow interception and sublimation. *Hydrological Processes*, *12*(15), 2317–2337. https://doi.org/10.1002/(SICI)1099-1085(199812)12:15%3C2317::AID-HYP799%3E3.0.CO;2-X

Pomeroy, J. W., & Schmidt, R. A. (1993). The use of fractal geometry in modelling intercepted snow accumulation and sublimation. *Eastern Snow Conference*, *50*, 231–239.

Raleigh, M. S., Gutmann, E. D., Van Stan II, J. T., Burns, S. P., Blanken, P. D., & Small, E. E. (2022). Challenges and capabilities in estimating snow mass intercepted in conifer canopies with tree sway monitoring. *Water Resources Research*, *58*(3). https://doi.org/10.1029/2021WR030972

Roesch, A., Wild, M., Gilgen, H., & Ohmura, A. (2001). A new snow cover fraction parameterization for the ECHAM4 GCM. *Climate Dynamics*, *17*(12), 933–946. https://doi.org/10.1007/s003820100153

Rojas-Heredia, F., Revuelto, J., Deschamps-Berger, C., Alonso-González, E., Domínguez-Aguilar, P., García, J., Pérez-Cabello, F., & López-Moreno, J. I. (2024). Snow depth distribution in canopy gaps in central pyrenees. *Hydrological Processes*, *38*(11), e15322. https://doi.org/10.1002/hyp.15322

Rutter, N., Essery, R., Pomeroy, J. W., Altimir, N., Andreadis, K. M., Baker, I., Barr, A., Bartlett, P., Boone, A., Deng, H., Douville, H., Dutra, E., Elder, K., Ellis, C. R., Feng, X., Gelfan, A., Goodbody, A., Gusev, Y., Gustafsson, D., … Yamazaki, T. (2009). Evaluation of forest snow processes models (SnowMIP2). *Journal of Geophysical Research: Atmospheres*, *114*(D6), 10–18. https://doi.org/10.1029/2008JD011063

Safa, H., Krogh, S. A., Greenberg, J., Kostadinov, T. S., & Harpold, A. A. (2021). Unraveling the controls on snow disappearance in montane conifer forests using multi-site lidar. *Water Resources Research*, *57*(12), 1–20. https://doi.org/10.1029/2020WR027522

Sanmiguel-Vallelado, A., López-Moreno, J. I., Morán-Tejeda, E., Alonso-González, E., Navarro-Serrano, F. M., Rico, I., & Camarero, J. J. (2020). Variable effects of forest canopies on snow processes in a valley of the central Spanish Pyrenees. *Hydrological Processes*, *34*(10), 2247–2262. https://doi.org/10.1002/hyp.13721

Sanmiguel-Vallelado, A., McPhee, J., Esmeralda Ojeda Carreño, P., Morán-Tejeda, E., Julio Camarero, J., & López-Moreno, J. I. (2022). Sensitivity of forest–snow interactions to climate forcing: Local variability in a Pyrenean valley. *Journal of Hydrology*, *605*. https://doi.org/10.1016/j.jhydrol.2021.127311

Schmidt, R. A., & Gluns, D. R. (1991). Snowfall interception on branches of three conifer species. *Canadian Journal of Forest Research*, *21*(8), 1262–1269. https://doi.org/10.1139/x91-176

Schmidt, R. A., & Pomeroy, J. W. (1990). Bending of a conifer branch at subfreezing temperatures: Implications for snow interception. *Canadian Journal of Forest Research*, *20*(8), 1251–1253. https://doi.org/10.1139/x90-165

Sicart, J. E., Pomeroy, J. W., Essery, R. L. H., & Bewley, D. (2006). Incoming longwave radiation to melting snow: Observations, sensitivity and estimation in Northern environments. *Hydrological Processes*, *20*(17), 3697–3708.

Staines, J., & Pomeroy, J. W. (2023). Influence of forest canopy structure and wind flow on patterns of sub-canopy snow accumulation in montane needleleaf forests. *Hydrological Processes*, *37*(10), 1–19. https://doi.org/10.1002/hyp.15005

Storck, P., Lettenmaier, D. P., & Bolton, S. M. (2002). Measurement of snow interception and canopy effects on snow accumulation and melt in a mountainous maritime climate, Oregon, United States. *Water Resources Research*, *38*(11), 1–16. https://doi.org/10.1029/2002wr001281

Szczypta, C., Gascoin, S., Houet, T., Hagolle, O., Dejoux, J.-F., Vigneau, C., & Fanise, P. (2015). Impact of climate and land cover changes on snow cover in a small Pyrenean catchment. *Journal of Hydrology*, *521*, 84–99. https://doi.org/10.1016/j.jhydrol.2014.11.060

Thackeray, C. W., Fletcher, C. G., & Derksen, C. (2014). The influence of canopy snow parameterizations on snow albedo feedback in boreal forest regions. *Journal of Geophysical Research: Atmospheres*, *119*(16), 9810–9821. https://doi.org/10.1002/2014JD021858

Troendle, C. A. (1983). The potential for water yield augmentation from forest management in the Rocky Mountain region. *Journal of the American Water Resources Association*, *19*(3), 359–373. https://doi.org/10.1111/j.1752-1688.1983.tb04593.x

Valante, F., David, J. S., & Gash, J. H. C. (1997). Modelling interception loss for two sparse eucalypt and pine forests in central Portugal using reformulated Rutter and Gash analytical models. *Journal of Hydrology*, *190*(1-2), 141–162. https://doi.org/10.1016/S0022-1694(96)03066-1

Verseghy, D. L. (2017). *CLASS – the Canadian Land Surface Scheme (version 3.6.1) technical documentation.* (p. 174) [Internal Report]. Environment and Climate Change Canada.

Viviroli, D., Kummu, M., Meybeck, M., Kallio, M., & Wada, Y. (2020). Increasing dependence of lowland populations on mountain water resources. *Nature Sustainability*, *3*(11), 917–928. https://doi.org/10.1038/s41893-020-0559-9

Weiskittel, A. R., Kershaw, J. A., Hofmeyer, P. V., & Seymour, R. S. (2009). Species differences in total and vertical distribution of branch- and tree-level leaf area for the five primary conifer species in Maine, USA. *Forest Ecology and Management*, *258*(7), 1695–1703. https://doi.org/10.1016/j.foreco.2009.07.035

Wheater, H. S., Pomeroy, J. W., Pietroniro, A., Davison, B., Elshamy, M., Yassin, F., Rokaya, P., Fayad, A., Tesemma, Z., Princz, D., Loukili, Y., DeBeer, C. M., Ireson, A. M., Razavi, S., Lindenschmidt, K.-E., Elshorbagy, A., MacDonald, M., Abdelhamed, M., Haghnegahdar, A., & Bahrami, A. (2022). Advances in modelling large river basins in cold regions with Modélisation Environmentale Communautaire—Surface and Hydrology (MESH), the Canadian hydrological land surface scheme. *Hydrological Processes*, *36*(4), 1–24. https://doi.org/10.1002/hyp.14557

Willmott, C. J. (1982). Some Comments on the Evaluation of Model Performance. *Bulletin of the American Meteorological Society*, *63*(11), 1309–1313. https://doi.org/10.1175/1520-0477(1982)063%3C1309:SCOTEO%3E2.0.CO;2

Zhong, F., Jiang, S., van Dijk, A. I. J. M., Ren, L., Schellekens, J., & Miralles, D. G. (2022). Revisiting large-scale interception patterns constrained by a synthesis of global experimental data. *Hydrology and Earth System Sciences*, *26*(21), 5647–5667. https://doi.org/10.5194/hess-26-5647-2022